# THE SOLVABILITY DICHOTOMY FOR COMMUTATOR-TREE MIXING IN FINITE GROUPS

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Abstract. Let $G$ be a finite group and $C \subseteq G$ a conjugation-closed generating set. Labelling the leaves of a balanced commutator tree by elements of $C$ and reading every expanded occurrence independently, as its element or the identity with probability $1/2$, produces a random product of $4^m$ independent lazy factors, the randomisation of Barrington's branching programs. We ask when every valid configuration, one whose deterministic commutator value is non-trivial, has a spectral gap in every non-trivial irreducible representation. For non-solvable $G$ such a uniform level exists if and only if every proper subgroup generated by elements of $C$ is solvable, and the first level is then the largest derived length of these subgroups. The proof combines a fixed-vector criterion, strict contraction being equivalent to generation by the leaves, with the normal-subset word-value lemma of Fernández-Alcober and Pintonello. Applied to every class, the criterion characterises Thompson's minimal simple groups, with universal level five attained by $\mathrm{PSL}_3(3)$. We classify involution classes of $\mathrm{PGL}_2(q)$ and transvection leaves in $\mathrm{SL}_n(q)$, lift the dichotomy to $\mathrm{GL}_2(\mathbb{Z}/p^r)$, and prove fixed-error mixing in optimal depth $\Theta(\log\log|G|)$ in the unobstructed families from the bounded-rank diameter theorem. For primes $q \equiv \pm 3 \pmod 8$, $q \neq 3$, every pair of exterior $S_4$ subgroups of $\mathrm{PGL}_2(q)$ meeting in an $S_3$ is simultaneously conjugate to a pair defined by Hamilton's quaternions, and the largest cosine between their fixed spaces over non-trivial irreducibles is at most $\sqrt{3}/2$, with asymptotic equality: it is the Ramanujan bound of the Lubotzky–Phillips–Sarnak graph $X^{3,q}$.

## 1. Introduction

Let $G$ be a finite group and let $C \subseteq G$ be a conjugation-closed set generating $G$, for instance a single conjugacy class. Label the $2^m$ leaves of a balanced binary tree of depth $m$ by elements of $C$ and let every internal vertex take the commutator $[x, y] = xyx^{-1}y^{-1}$ of its two children. Expanding the root value as a word in the leaf labels gives a product of $4^m$ occurrences, each a leaf label or its inverse. Now read every occurrence independently: with probability $1/2$ it contributes its element, otherwise the identity. The output is a random product of $4^m$ independent lazy factors, that is, a lazy random walk on $G$ whose steps are prescribed by the tree rather than chosen freely. We ask when every such walk contracts towards the uniform distribution, and what determines the depth at which this begins.

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Two features distinguish the construction from a word map. First, repeated occurrences of a label, including the inverse copies inside each commutator, receive independent coin tosses, so the output is not the value of a word at random arguments; in particular it is not confined to the derived subgroup, and one-dimensional characters are contracted by the individual lazy factors. Second, only *valid* configurations are considered, those whose deterministic commutator value at the root is non-trivial. This constraint is what brings the derived series into the problem: a configuration whose leaves all lie in a solvable subgroup $H$ has trivial root value as soon as the depth reaches the derived length of $H$.

We call a labelled tree a *configuration*, and say that it *contracts strictly* if its Fourier operator has norm less than one in every non-trivial irreducible representation of $G$, that is, if the walk has a spectral gap on the orthogonal complement of the constants. A subgroup of $G$ is *$C$-generated* if it is generated by its intersection with $C$. Precise definitions are given in Section 2.

**The solvability dichotomy.** The answer is governed by solvability, and the depth by derived length.

**Theorem A** (Solvability dichotomy and exact level)**.** *Let $G$ be a finite non-solvable group and $C$ a conjugation-closed generating set of $G$.*

(1) *If some proper $C$-generated subgroup of $G$ is non-solvable, then at every depth there is a valid configuration that fails to contract in some non-trivial irreducible representation.*
(2) *If every proper $C$-generated subgroup of $G$ is solvable, let $D$ be the largest derived length of such a subgroup. Then every valid configuration of depth at least $D$ contracts strictly, and at every depth less than $D$ some valid configuration does not.*

Thus $(G, C)$ has a *uniform strict-contraction level*, a depth from which every valid configuration contracts strictly, if and only if every proper $C$-generated subgroup is solvable, and the level is then exactly $D$ (Theorem 3.7). Equivalently, $D$ is the exact depth from which a non-trivial iterated commutator certifies that its entries generate $G$. Transpositions illustrate both cases. In $S_5$ every proper subgroup generated by transpositions is solvable, the largest derived length is that of $S_4$, and the level is 3. In $S_n$ with $n \geq 6$ the point stabiliser $S_{n-1}$ is a proper non-solvable subgroup generated by transpositions, and at every depth it supplies valid configurations supported in a proper subgroup (Corollary 4.1). Only $C$-generated subgroups matter: for the involutions of $A_5$ the level is 2, because the involutions of $A_4$ generate only $V_4$ (Corollary 4.2). Table 1 lists further levels.

Applied to all conjugacy classes at once, Theorem A characterises the minimal simple groups classified by Thompson [59].

**Corollary 1.1** (Minimal simple groups)**.** *A finite non-abelian simple group $G$ is minimal simple, that is, all of its proper subgroups are solvable, if and only if every non-trivial conjugacy class of $G$ has a uniform strict-contraction level. In that case every class has level at most* 5*, and the involution class of* $\mathrm{PSL}_3(3)$ *has level exactly* 5 (Corollary 4.3 and Proposition 4.4).

| Group and leaf set $C$ | Decisive proper $C$-generated subgroups | Level |
|---|---|---|
| $S_5$, transpositions | $S_4$, of derived length 3 | 3 |
| $S_n$, transpositions, $n \geq 6$ | $S_{n-1}$, non-solvable | none |
| $A_5$, involutions | $S_3$ and $D_{10}$, of derived length 2 | 2 |
| $\mathrm{PSL}_2(7)$, involutions | $S_4$ | 3 |
| $\mathrm{PSL}_3(3)$, involutions | $3^2{:}\,\mathrm{GL}_2(3)$, of derived length 5 | 5 |
| $\mathrm{SL}_3(3)$, transvections | $3^2{:}\,\mathrm{SL}_2(3)$, of derived length 4 | 4 |
| $\mathrm{PGL}_2(q)$, all involutions | $\mathrm{PSL}_2(q)$, non-solvable | none |
| $\mathrm{PGL}_2(q)$, exterior involutions | subfield subgroups of odd degree; otherwise affine, dihedral or $S_4$ | 2, 3 or none (Theorem C) |

TABLE 1. Uniform strict-contraction levels for some leaf sets; $q \geq 5$ is odd (Sections 4 and 5).

The proof of Theorem A has two ingredients. The first is elementary. The Fourier operator of a configuration is a product of lazy averages $\frac{1}{2}(I + \rho(g))$, and such a product has norm one exactly when the leaves have a common non-zero fixed vector; hence a configuration contracts strictly if and only if its leaves generate $G$ (Theorem 3.2). The second is the normal-subset word-value lemma of Fernández-Alcober and Pintonello [24, Lemma 2.1(ii)]: for a conjugation-closed generating subset of a group $H$, the depth-$k$ commutator values of its elements generate $H^{(k)}$. Together they show that validity forces the leaves out of every proper solvable $C$-generated subgroup once the depth reaches its derived length, and never forces them out of a non-solvable one. Contraction has to be tested in every irreducible representation: a proper subgroup that is transitive in some permutation action is invisible to the corresponding permutation representation (Remark 3.10).

Theorem A is related to two lines of work. Probabilistic generation asks how often a random tuple escapes every proper subgroup: for finite simple groups almost always, by Dixon, Kantor–Lubotzky and Liebeck–Shalev [22, 35, 44], with refinements for elements of prescribed type [18, 45] and beyond simple groups [31]. Here the escape is forced by validity rather than by chance, and the derived length records when. Secondly, solvability is detected through subgroups generated by few or prescribed elements in Thompson's classification [59], in the descriptions of the solvable radical by two-generated subgroups and by conjugates [27, 30], in two-variable laws [6, 13] and in probabilistic identities [42]. Theorem A is a statement of this kind with an analytic conclusion: solvability of the proper $C$-generated subgroups is equivalent to the eventual onset of a uniform spectral gap, and their derived lengths give the exact onset.

**A sharp angle between octahedral subgroups.** When no common fixed vector exists, contraction is governed by angles between fixed spaces. For subgroups $A, B \leq G$ and a representation $\rho$, let $P_A^\rho$, $P_B^\rho$ be the orthogonal projections onto the fixed spaces; then $\|P_A^\rho P_B^\rho\|$ is the cosine of the angle between these spaces, and for conjugate subgroups it is the norm of a double coset in a Hecke algebra (Lemma 6.1).

Such angles control alternating projections [5, 38] and give Kazhdan constants from arrangements of subgroups [36]. Our main concrete result computes one of them sharply, uniformly in a family of rank one.

For a prime $q \equiv \pm 3 \pmod 8$ the subgroups of $\mathrm{PGL}_2(q)$ isomorphic to $S_4$ are not contained in $\mathrm{PSL}_2(q)$; we call them *exterior octahedral* subgroups. They form a single conjugacy class [26, 34, 40]. Reducing Hamilton's quaternions modulo $q$ produces an explicit such subgroup $A_q$ and a conjugate $B_q = hA_qh^{-1}$, where $h = i + j + k$.

**Theorem B** (The natural octahedral pair)**.** *Let $q \neq 3$ be a prime with $q \equiv \pm 3 \pmod 8$ and put $G_q = \mathrm{PGL}_2(q)$. Then $A_q \cap B_q \cong S_3$ and $\langle A_q, B_q\rangle = G_q$. Every pair $A_1, A_2$ of exterior octahedral subgroups of $G_q$ with $|A_1 \cap A_2| = 6$ is simultaneously conjugate to $(A_q, B_q)$, and*

$$\max_{\rho\in\widehat{G_q},\,\rho\neq\mathbf{1}} \|P^{\rho}_{A_1}P^{\rho}_{A_2}\| = c_q \le \frac{\sqrt{3}}{2}, \qquad c_q \longrightarrow \frac{\sqrt{3}}{2} \quad (q\to\infty),$$

*where $c_q$ is the value attained by the pair $(A_q, B_q)$* (Proposition 6.3 and Theorem 6.6).

The mechanism is a compression identity. The four quaternions $\pm i \pm j \pm k$ of norm 3 represent the double coset $A_qhA_q$, so the Hecke operator of the 4-regular Lubotzky–Phillips–Sarnak graph [49] $X^{3,q}$, restricted to the $A_q$-fixed vectors of each irreducible representation, *is* the angle operator $P_{A_q}\rho(h)P_{A_q}$. The Ramanujan bound $2\sqrt{3}$, divided by the degree 4, is $\sqrt{3}/2 = \cos(\pi/6)$, and the Alon–Boppana bound on the quotient graphs gives optimality. Geometrically, the smallest angle between the two fixed spaces over all non-trivial irreducible representations is at least $\pi/6$ and tends to $\pi/6$. Combined with an estimate inside $S_4$, the angle bound yields an explicit uniform contraction constant for the corresponding configurations (Corollary 6.8).

By Proposition 6.3 two distinct exterior octahedral subgroups meet in a subgroup of order at most 6, so Theorem B settles the largest intersection in the following problem, which was raised in the preliminary preprint [62, Conjecture 6.9].

**Conjecture 1.2** (Uniform angle for exterior octahedral pairs)**.** *For primes $q \equiv \pm 3 \pmod 8$, $q \neq 3$,*

$$\sup_{\substack{A,B\cong S_4 \text{ exterior}\\ \langle A,B\rangle=\mathrm{PGL}_2(q)}} \max_{\rho\neq\mathbf{1}} \|P^{\rho}_{A}P^{\rho}_{B}\| \le \frac{\sqrt{3}}{2} + o(1) \qquad (q\to\infty).$$

The intersection orders $1, 2, 3, 4$ remain open; numerical all-pairs maxima for five primes are recorded in Table 2, and a qualitative bound $1 - b$ with an absolute $b > 0$ holds outside a set of primes of relative density zero (Corollary 7.4).

**Exact levels in rank one.** The subgroup criterion can be evaluated in infinite families. An involution of $\mathrm{PGL}_2(q)$ is *exterior* if it lies outside $\mathrm{PSL}_2(q)$.

**Theorem C** (Rank-one involutions)**.** *Let $q \ge 5$ be odd. In $\mathrm{PGL}_2(q)$ the set of all involutions has no uniform strict-contraction level. The exterior class has a level if*

*and only if* $\mathbb{F}_q$ *has no proper subfield* $\mathbb{F}_{q_0}$ *with* $q_0 \geq 5$ *and* $[\mathbb{F}_q : \mathbb{F}_{q_0}]$ *odd; the level is then* 3 *for* $q \equiv \pm 3 \pmod 8$ *and* 2 *for* $q \equiv \pm 1 \pmod 8$. *The involution class of* $\mathrm{PSL}_2(q)$ *is classified in Theorem 5.2.*

The level depends on the class and not only on the group: in $\mathrm{PSL}_3(3)$ the involutions have level 5 and the transvections level 4. For transvections in $\mathrm{SL}_n(q)$, $n \geq 3$, the level is 3 for $\mathrm{SL}_3(2)$ and 4 for $\mathrm{SL}_3(3)$, and in every other case a proper non-solvable subgroup generated by transvections obstructs (Proposition 4.6). For a generating conjugacy class of $\mathrm{GL}_2(\mathbb{Z}/p^r)$, $p \geq 5$, the dichotomy is inherited from the reduction modulo $p$, and at $r = 2$ the level increases by at most one (Proposition 4.8).

**Rates.** For a fixed group, strict contraction gives a positive gap by finiteness; in a family of groups the gap may tend to zero. Write $U_G$ for the uniform distribution on $G$ and $\mu_\tau$ for the output distribution of a configuration $\tau$.

**Theorem D** (Uniform rates and optimal order of depth)**.** *For the classes with a level in Theorem C and Theorem 5.2, and for every non-trivial conjugacy class of every minimal simple group, there are absolute constants* $c, B > 0$ *such that every valid configuration* $\tau$ *of depth* $m \geq D$ *satisfies*

$$d_{\mathrm{TV}}(\mu_\tau, U_G) \leq \tfrac{1}{2}\sqrt{|G| - 1}\,\exp\bigl(-c4^m/(\log|G|)^B\bigr).$$

*For a fixed error in* $(0, 1/2)$*, the least depth guaranteeing this error for all valid configurations has order* $\Theta(\log\log|G|)$ *as* $|G| \to \infty$ (Theorem 7.1, Proposition 7.2, and Corollary 7.3).

The upper bound converts the bounded-rank diameter theorem of Pyber–Szabó [53] into a contraction estimate at depth $D$ and amplifies it through the tree; the lower bound holds because the output at depth $m$ is supported on at most $2^{4^m}$ elements. For involution leaves at bounded depth, a uniform contraction gap is equivalent to a uniform spectral gap for the associated lazy Cayley walks (Corollary 3.15), so the quantitative problem is one of expansion for particular generating sets, in the setting of quasirandomness, product growth and the Bourgain–Gamburd method [4, 10–12, 16, 17, 28, 32, 37, 55]; bounded derived length is the corresponding obstruction in solvable families [50]. Accordingly, over prime fields outside a set of relative density zero, the uniform expansion theorem of Breuillard–Gamburd [14] improves the rate to $e^{-c4^m}$ (Corollary 7.4), and over arbitrary finite fields the theorem of Becker–Breuillard [8] bounds the number of exceptional configurations at each fixed depth, up to simultaneous conjugacy, by $O(q^\delta)$ for every $\delta > 0$ (Corollary 7.5).

**Related work.** Word maps relate random arguments to the distribution of word values [15, 41, 58]. Every element of a finite non-abelian simple group is a commutator [47]; iterated commutators at uniform arguments are asymptotically equidistributed on simple groups of growing order [25]; the walk whose steps are uniform values of a fixed word mixes in boundedly many steps [56]; and products of word values in disjoint variables are almost uniform [43]. These results concern a fixed word at uniform arguments in simple groups of large order. Here the arguments come from a prescribed conjugation-closed set, repeated occurrences are

read independently (Remark 2.3), and the group is fixed while the depth grows; the question is which subgroups obstruct contraction and at what depth the obstruction disappears.

When $C$ is a single class, the walk with steps uniform in $C$ is diagonalised by character ratios, and its mixing is classical [20, 21, 33, 46, 51, 54, 57]. In the commutator tree the steps are prescribed, and trapping is detected through fixed vectors in induced representations rather than through character bounds. In a different direction, Amir, Blachar, Ghosh and Vishne [1] ask when some finite product of lazy steps, with freely chosen elements and probabilities, is exactly uniform; every $S_n$ has this property [2]. We fix fair coins and the commutator pattern and ask for contraction uniformly over valid configurations: among the groups $S_n$, $n \geq 5$, with transposition leaves, only $S_5$ has a level.

**Origin.** The construction is Barrington's [7]: width-five branching programs over $S_5$ compute $\mathrm{NC}^1$ by combining their instructions through a balanced tree of commutators, and over matrix groups one may use pairs conjugate to their commutator [9]. Group programs of this kind are randomised in secure computation [39], and the distribution of products of random group elements underlies leakage-resilient circuits over $A_5$ [52] and the interleaved-product bounds of Gowers–Viola [29]. The present question is what happens when each instruction of the expanded program is independently replaced by the identity with probability $1/2$. The companion paper [63] treats $S_5$ quantitatively; here we establish the general structural law. The reduction from the conditioned laws of branching programs to the walk studied here is recalled in Section 2.1.

**Organisation.** Section 2 sets up the model. Section 3 proves the fixed-vector criterion, Theorem A and the quantitative stability estimates used later. Section 4 computes levels for symmetric and alternating groups, minimal simple groups, transvections and congruence quotients, and proves Corollary 1.1. Section 5 proves Theorem C, Section 6 proves Theorem B, and Section 7 proves Theorem D. An extended version of this paper [61] contains additional material that is not needed here: the localisation of obstructions under quotients, the levels attained by prescribed families of configurations, and further computational tables.

## 2. Preliminaries

**Configurations.** Let $G$ be a finite group. A *configuration*, or *tree*, of *level* $k \geq 0$ is a balanced binary tree with $2^k$ leaves, each labelled by an element of $G$. We write $\tau = [\tau_1, \tau_2]$ for the configuration of level $k+1$ with left and right subtrees $\tau_1, \tau_2$, and call the subtrees of a configuration its *sub-blocks*, or simply *blocks*. The *target* $\alpha(\tau) \in G$ and the *expanded word* $w(\tau)$, a finite sequence of elements of $G$, are defined recursively. A leaf labelled $\sigma$ has $\alpha = \sigma$ and $w = (\sigma)$, and

$$\alpha([\tau_1, \tau_2]) = [\alpha(\tau_1), \alpha(\tau_2)], \qquad w([\tau_1, \tau_2]) = w(\tau_1)\, w(\tau_2)\, w(\tau_1)^{-1}\, w(\tau_2)^{-1},$$

where $[x, y] = xyx^{-1}y^{-1}$, words are concatenated, and the inverse of a word $(g_1, \dots, g_n)$ is $(g_n^{-1}, \dots, g_1^{-1})$. Thus $w(\tau)$ has $4^k$ letters, each a leaf label or its inverse, and their product is $\alpha(\tau)$. The configuration is *valid* if $\alpha(\tau) \neq e$, and

*involutive* if all its leaf labels are involutions. Its *leaf subgroup* $H(\tau)$ is the subgroup generated by its leaf labels.

**The independent-read model.** Let $w(\tau) = (g_1, \dots, g_N)$ with $N = 4^k$, and let $\varepsilon_1, \dots, \varepsilon_N$ be independent uniform bits. The *output law* $\mu_\tau$ is the distribution of

$$g_1^{\varepsilon_1} g_2^{\varepsilon_2} \cdots g_N^{\varepsilon_N}.$$

Every occurrence is thus read independently, including the occurrences inside the inverse copies $w(\tau_i)^{-1}$. The terminology comes from branching programs; see Section 2.1.

**Notation.** Representations are unitary, with carrier space $V_\rho$ and dimension $d_\rho$. We write $\widehat{G}$ for the irreducible representations and $\mathbf{1}$ for the trivial one. For a probability measure $\mu$, put $\widehat{\mu}(\rho) = \sum_g \mu(g)\rho(g)$ and $M_\tau(\rho) = \widehat{\mu_\tau}(\rho)$. The notation $\|v\|$ denotes the Hilbert-space norm and $\|A\|$ the operator norm; $\|A\|_{\mathrm{F}}$ denotes the Frobenius norm. We use $V_\rho^H$ for the $H$-fixed subspace and $\mathrm{Fix}_\rho(s) = V_\rho^{\langle s \rangle}$ for a single element. When $\rho$ is fixed, we may write $V = V_\rho$ and $V^H = V_\rho^H$. For $w \in G$, its conjugacy class is denoted by $w^G$.

**Contraction and levels.** A configuration $\tau$ *contracts strictly* if $\|M_\tau(\rho)\| < 1$ for every non-trivial irreducible representation $\rho$ of $G$. Let $C \subseteq G$ be closed under conjugation. A subgroup is *$C$-generated* if it is generated by its intersection with $C$. The *uniform strict-contraction level* of $(G, C)$, when it exists, is the least level at which every valid configuration with leaves in $C$ contracts strictly; by Theorem 3.7 every valid configuration of larger level then contracts strictly as well.

**Lemma 2.1** (Independent-read factorisation)**.** *Let $w(\tau) = (g_1, \dots, g_N)$. For every unitary representation $\rho$,*

$$M_\tau(\rho) = P_{g_1}(\rho) \cdots P_{g_N}(\rho), \qquad P_\sigma(\rho) = \tfrac{1}{2}\big(I + \rho(\sigma)\big).$$

*We call $P_\sigma(\rho)$ the* leaf operator *of $\sigma$. If $\sigma$ is an involution, it is the orthogonal projection onto* $\mathrm{Fix}_\rho(\sigma)$.

*Proof.* The expectation of $\rho(g_1^{\varepsilon_1}) \cdots \rho(g_N^{\varepsilon_N})$ factors by independence, and a uniform bit $\varepsilon$ gives $\mathbb{E}\,\rho(g^\varepsilon) = \frac{1}{2}(I + \rho(g))$. If $\sigma^2 = e$ then $\rho(\sigma)$ is a self-adjoint unitary involution, so $P_\sigma$ is self-adjoint and $P_\sigma^2 = \frac{1}{4}(I+\rho(\sigma))^2 = \frac{1}{4}(2I+2\rho(\sigma)) = P_\sigma$. □

**Lemma 2.2** (Recursive commutator-tree operator)**.** *Let $\tau = [\tau_1, \tau_2]$. Fix $\rho$ and put $X = M_{\tau_1}(\rho)$ and $Y = M_{\tau_2}(\rho)$. Then*

$$M_\tau(\rho) = XYX^\dagger Y^\dagger.$$

*For a single leaf $s$, the operator is $P_s = P_s(\rho)$. Thus a node with leaves $s, t$ has operator $P_s P_t P_s^\dagger P_t^\dagger$, equal to $(P_s P_t)^2$ when both leaves are involutions. Expanding the recursion gives a product of leaf averages and their adjoints; involution leaf averages are orthogonal projections.*

*Proof.* The four segments of $w(\tau)$ use disjoint sets of bits, so their outputs $a, b, a', b'$ are independent, and $a', b'$ are distributed as $a^{-1}, b^{-1}$. Unitarity gives $\mathbb{E}\rho(a') = X^\dagger$

and $\mathbb{E}\rho(b') = Y^\dagger$. Independence therefore yields

$$\mathbb{E}[\rho(a)\rho(b)\rho(a')\rho(b')] = XYX^\dagger Y^\dagger.$$

For involution leaves, $P_s^\dagger = P_s$ and $P_t^\dagger = P_t$ by Lemma 2.1. □

*Remark* 2.3 (Independent occurrences and word evaluation). The two distributions

$$\underbrace{XYX^{-1}Y^{-1}}_{\text{same } X,Y \text{ (correlated)}} \qquad \text{and} \qquad \underbrace{XY\,\tilde{X}^{-1}\tilde{Y}^{-1}}_{\tilde{X},\tilde{Y} \text{ independent copies of } X,Y}$$

encode different dependence between repeated occurrences. Ordinary word evaluation uses the same value of $X$ in $X$ and $X^{-1}$. The independent-read model uses four independent blocks, so its Fourier transform factors as in Lemma 2.2. This factorisation gives the fixed-vector criterion of Lemma 3.1.

**Mixing bound.** The Diaconis–Shahshahani lemma [21] gives

$$4\, d_{\mathrm{TV}}(\mu, U_G)^2 \le \sum_{\rho \neq \mathbf{1}} d_\rho \|\widehat{\mu}(\rho)\|_{\mathrm{F}}^2,$$

the sum over non-trivial irreducibles. Since $\|M\|_{\mathrm{F}}^2 \le d_\rho \|M\|^2$, controlling $\|M_\tau(\rho)\|$ for every non-trivial $\rho$ controls $d_{\mathrm{TV}}(\mu_\tau, U_G)$. Every total-variation upper bound below may also be capped at 1.

2.1. **Branching programs and the conditioned law.** The model originates in Barrington's theorem [7]. A *permutation branching program* of length $L$ over $G$ is a sequence of instructions $(i_j, g_j^{(0)}, g_j^{(1)})$, $1 \le j \le L$; on input $x \in \{0,1\}^\ell$ it outputs $\prod_{j=1}^{L} g_j^{(x_{i_j})}$. Barrington's program for the point function at $T \in \{0,1\}^\ell$ is built from a *template*, a configuration $\tau$ together with a read sequence $(i_j)$: the $j$-th instruction outputs the $j$-th letter of $w(\tau)$ when the bit it reads agrees with $T$, and the identity otherwise. If the $L = 4^k$ reads are replaced by independent uniform bits, the output has law $P_T = \mu_\tau$, which does not depend on $T$. Genuine inputs correspond to *consistent* reads, in which all instructions reading the same variable see the same bit. Conditioning $P_T$ on consistent or on inconsistent reads gives laws $C_T$ and $D_T$. The following reductions from [63, Theorem 3 and Lemma 1] transfer mixing estimates for $\mu_\tau$ to $D_T$; the rest of the paper concerns $\mu_\tau$ only.

**Theorem 2.4** ($T$-independence)**.** *Let* $\mathrm{BP}_T, \mathrm{BP}_{T'}$ *be point-function programs built from the same template (same read sequence, same leaf element* $\sigma_j$ *at each leaf), differing only in which branch is the matching one at the literals where* $T_i \neq T_i'$*. Then* $D_T = D_{T'}$ *over any group* $G$*.*

*Proof.* Let $S = \{i : T_i \neq T_i'\}$ and define $F_S : \{0,1\}^L \to \{0,1\}^L$ flipping $b_j$ exactly when the literal read at step $j$ lies in $S$. Then (i) $F_S$ is an involution; (ii) it flips all readings of each $S$-variable simultaneously, hence preserves the consistency/inconsistency of each variable individually; and (iii) at every step the swap of the matching branch ($T \leftrightarrow T'$) and the bit flip cancel, so the output is unchanged: $\varphi_{T'}(b) = \varphi_T(F_S(b))$. As $F_S$ is a measure-preserving involution of the inconsistent paths, change of variables gives $D_{T'} = D_T$. □

The same bijection proves $T$-independence over every finite group and extends to programs with a fixed wildcard pattern.

**Lemma 2.5** (Conditioning)**.** *Assume $0 < p = \Pr[\textit{consistent}] < 1$, so both conditional laws are defined. Then $P_T = (1-p)D_T + p\,C_T$, hence*

$$d_{\mathrm{TV}}(D_T, U_G) \;\le\; d_{\mathrm{TV}}(P_T, U_G) + p \cdot d_{\mathrm{TV}}(D_T, C_T) \;\le\; d_{\mathrm{TV}}(P_T, U_G) + p.$$

*If every variable is read at least once then the consistent paths are in bijection with $\{0,1\}^{\ell}$, so $p = 2^{\ell}/2^{L}$.*

*Proof.* The first identity is the law of total probability over the consistency event; the triangle inequality and $d_{\mathrm{TV}} \le 1$ give the bound. The count of consistent paths is the number of per-variable constant assignments. □

Any bound on $d_{\mathrm{TV}}(P_T, U_G)$ therefore gives a bound for $D_T$ with additive loss $p = 2^{\ell - L}$. For $L = \ell^2$, this loss is $2^{-\Theta(\ell^2)}$.

2.2. **Computer-assisted statements.** Three statements below depend on machine computation. The constant $17/200$ in the proof of Corollary 6.8 is certified in exact rational arithmetic. The exact values of $c_q$ in Section 6.5 are obtained by factoring integer characteristic polynomials. The entries of Table 2 are floating-point values, offered as evidence and not used in any proof. All other results are proved by hand; the artefact nevertheless contains scripts that independently recompute the subgroup data used in Propositions 4.4, 4.6 and 6.3 and several of the levels in Table 1. The computations were run with Python 3.9.6, NumPy 2.0.2 and SymPy 1.14.0. The scripts, their outputs and the extended version of this paper are archived at [61].

## 3. The fixed-vector criterion and the solvability dichotomy

### 3.1. When does an operator fail to contract?

**Lemma 3.1** (Geometric lemma)**.** *Fix a configuration $\tau$ with leaf set $S \subseteq G$ and let $H = \langle S \rangle$. For any unitary representation $\rho$ on $V$,*

$$\|M_\tau(\rho)\| = 1 \quad \Longleftrightarrow \quad V^H \neq 0.$$

*Proof.* Each leaf operator $P_\sigma(\rho) = \frac{1}{2}(I + \rho(\sigma))$ is a contraction: $\rho(\sigma)$ is unitary with eigenvalues $e^{i\theta}$, so $P_\sigma(\rho)$ has eigenvalues $\frac{1}{2}(1+e^{i\theta})$ of modulus $|\cos(\theta/2)| \le 1$, equal to 1 only on the $+1$-eigenspace of $\rho(\sigma)$, i.e. on $\mathrm{Fix}_\rho(\sigma)$. By Lemma 2.2, $M_\tau(\rho)$ is a product $M_1 \cdots M_N$ in which each $M_t$ is a leaf operator $P_\sigma(\rho)$ or its adjoint $P_\sigma(\rho)^\dagger = P_{\sigma^{-1}}(\rho)$ (some $\sigma \in S$); both are normal contractions whose norm-1 eigenspace is exactly $\mathrm{Fix}_\rho(\sigma) = \mathrm{Fix}_\rho(\sigma^{-1})$. If $\|M_1 \cdots M_N v\| = \|v\|$ for a unit vector $v$, put $v_N = v$ and $v_{t-1} = M_t v_t$. Since every $M_t$ is a contraction and $\|v_0\| = \|v_N\|$, all inequalities $\|v_{t-1}\| \le \|v_t\|$ are equalities. Equality for $M_t$ forces $v_t \in \mathrm{Fix}_\rho(\sigma_t)$, so $M_t v_t = v_t$ and $v_{t-1} = v_t$. Inducting gives $v_t = v$ for all $t$, hence $v$ is fixed by every leaf, i.e. lies in $\bigcap_{\sigma \in S} \mathrm{Fix}_\rho(\sigma) = V^H$. Conversely any $v \in V^H$ is fixed by $M_\tau(\rho)$. As $\|M_\tau(\rho)\| \le 1$ always, the claim follows. □

**Theorem 3.2** (Fixed-vector criterion)**.** *The Fourier operator of a configuration $\tau$ contracts strictly in* every *non-trivial irreducible representation* if and only if its leaves generate $G$*:*

$$\|M_\tau(\rho)\| < 1 \ \ \forall\, \rho \neq \mathbf{1} \quad \Longleftrightarrow \quad H(\tau) = G.$$

*Proof.* Put $H = H(\tau)$. By Lemma 3.1 the left side fails for some non-trivial $\rho$ iff $V_\rho^H \neq 0$ for some non-trivial $\rho$, i.e. iff $\mathrm{Ind}_H^G \mathbf{1} = \mathbb{C}[G/H]$ contains a non-trivial irreducible. Now $\mathbb{C}[G/H] = \bigoplus_\rho (\dim V_\rho^H)\,\rho$ has dimension $[G : H]$ and contains the trivial representation exactly once; it contains a non-trivial irreducible iff $[G : H] > 1$, i.e. iff $H \neq G$. Hence all non-trivial $\rho$ contract iff $H = G$. □

*Remark* 3.3. Combined with submultiplicativity $\|ABA^\dagger B^\dagger\| \leq \|A\|^2\|B\|^2 \leq \max\{\|A\|, \|B\|\}^4$ and the Diaconis–Shahshahani bound, Theorem 3.2 says: in a fixed finite instance, once every sub-block at some level $k_0$ has leaves generating $G$, every non-trivial Fourier block is strictly contractive; taking the maximum over the finitely many sub-blocks and irreducibles gives a gap $< 1$, and further levels then decay. For a family of groups a quantitative total-variation bound also needs an explicit lower bound on this gap; this is the subject of Section 7.

### 3.2. **From the criterion to a solvability dichotomy.**

**Lemma 3.4** (Validity is one-directional)**.** *The level-$k$ commutator target of a configuration lies in $H^{(k)}$, the $k$-th derived subgroup of $H = \langle S\rangle$ (the leaf subgroup). In particular, if $H^{(k)} = \{e\}$ the target is trivial and the configuration is* invalid*.*

*Proof.* Induct on $k$. A level-0 target is a single leaf $\sigma \in S \subseteq H = H^{(0)}$. A level-$(k{+}1)$ target is a commutator $[A, B]$ of two level-$k$ targets $A, B \in H^{(k)}$ (inductive hypothesis); hence $[A, B] \in [H^{(k)}, H^{(k)}] = H^{(k+1)}$. The converse fails: $H^{(k)} \neq \{e\}$ does not force a specific target to be non-trivial. □

By Theorem 3.2 a contraction failure (in some irreducible) at level $k$ requires a *proper* leaf subgroup $H \neq G$ with $H^{(k)} \neq \{e\}$. To turn this into a dichotomy we fix a *conjugation-closed* leaf set $C$ (a union of conjugacy classes of $G$); all our examples—transpositions in $S_n$, involutions in $A_5$ or $\mathrm{PGL}_2$—are of this form. The key fact is that, ranging over such a set, the realisable targets generate the terms of the derived series.

**Lemma 3.5** (Targets generate the derived series; cf. [24])**.** *Let $C \subseteq G$ be closed under conjugation by $H := \langle C\rangle$. Define the realisable level-$k$ targets by $T_0 = C$ and $T_{k+1} = \{[a, b] : a, b \in T_k\}$, where $a, b$ range independently (the two sub-blocks of a node may be labelled independently). Then $\langle T_k\rangle = H^{(k)}$ for every $k \geq 0$.*

*Proof.* Induct on $k$; the base case is $\langle C\rangle = H = H^{(0)}$. Conjugation by $g \in H$ sends $[a, b] \mapsto [a^g, b^g]$ with $a^g, b^g \in T_k$, since $C$—and hence each $T_k$—is closed under $H$-conjugation; thus $T_{k+1}$ is $H$-conjugation-closed and $\langle T_{k+1}\rangle$ is normal in $H$. Write $M = H^{(k)} = \langle T_k\rangle$. Each $[a, b]$ $(a, b \in T_k \subseteq M)$ lies in $[M, M] = H^{(k+1)}$, so $\langle T_{k+1}\rangle \subseteq H^{(k+1)}$. Conversely, $[M, M]$ is the normal closure in $M$ of $\{[s, t] : s, t \in T_k\}$ (in $M/N$, with $N$ that normal closure, the images of

the generating set $T_k$ commute, so $M/N$ is abelian); as $\langle T_{k+1}\rangle$ is normal in $M$ and contains those commutators, it contains $[M, M] = H^{(k+1)}$. Hence equality. □

**Definition 3.6** ($C$-generated subgroup)**.** A subgroup $H \leq G$ is *$C$-generated* if $H = \langle H \cap C\rangle$. Every leaf subgroup $L = \langle S\rangle$ with $S \subseteq C$ is $C$-generated; and, $C$ being conjugation-closed in $G$, $L \cap C$ is closed under conjugation by $L$.

**Theorem 3.7** (Solvability dichotomy and exact level)**.** *Let $G$ be a finite non-solvable group and $C \subseteq G$ a conjugation-closed set with $\langle C\rangle = G$.*

*(1)* ***(Obstruction.)*** *If some proper $C$-generated subgroup is non-solvable, then for every level $k$ a valid configuration $\tau$ has $\|M_\tau(\rho)\| = 1$ for some non-trivial $\rho$: the leaf set has* no uniform strict-contraction level.

*(2)* ***(Sharp level.)*** *If every proper $C$-generated subgroup is solvable, put $D = \max\{\mathrm{dl}(H) : H < G,\ H \text{ is } C\text{-generated}\}$. Then every valid configuration of level $\geq D$ has leaf subgroup $G$ and contracts strictly in all non-trivial irreducibles, while some valid configuration of level $D-1$ does not. The level of strict contraction is exactly $D$.*

*Proof. (1)* Let $H < G$ be proper, $C$-generated and non-solvable; its derived series stabilises at a non-trivial perfect core, so $H^{(k)} \neq \{e\}$ for all $k$. By Lemma 3.5 inside $H$ (with the $H$-conjugation-closed set $H \cap C$, which generates $H$ by Definition 3.6), the level-$k$ targets with leaves in $H \cap C$ generate $H^{(k)} \neq \{e\}$, so a valid level-$k$ configuration $\tau$ with leaves in $H$ exists. Its leaf subgroup is $\leq H < G$, so by Theorem 3.2 some non-trivial $\rho$ has $V_\rho^{H(\tau)} \neq 0$, whence $\|M_\tau(\rho)\| = 1$ (Lemma 3.1). *(2)* If a valid level-$k$ configuration ($k \geq D$) had leaf subgroup $L \neq G$, then $L$ is proper, $C$-generated and, by hypothesis, solvable with $\mathrm{dl}(L) \leq D \leq k$, so its target lies in $L^{(k)} \subseteq L^{(D)} = \{e\}$ (Lemma 3.4)—contradicting validity. Thus $L = G$ and it contracts strictly (Theorem 3.2). For exactness, take proper $C$-generated $H$ with $\mathrm{dl}(H) = D$; by Lemma 3.5 inside $H$ the level-$(D-1)$ targets generate $H^{(D-1)} \neq \{e\}$, giving a valid level-$(D-1)$ configuration with leaf subgroup $\leq H < G$ that fails to contract (Lemma 3.1, Theorem 3.2). □

The contraction level is therefore exactly the largest derived length among proper $C$-generated subgroups, and yields a quantitative bound for each fixed group. The value $D$ is also the first level at which uniform contraction holds: if a proper $C$-generated $H$ has $\mathrm{dl}(H) = D$, then $H^{(k)} \neq \{e\}$ for every $k < D$, and Lemma 3.5 inside $H$ supplies a valid configuration with leaf subgroup contained in $H$ at each of these levels, so strict contraction fails at every level below $D$.

**Proposition 3.8** (Quantitative bound, fixed group)**.** *In case (2) of Theorem 3.7, let $\lambda = \lambda_{G,C} < 1$ be the maximum of $\|M_\tau(\rho)\|$ over non-trivial irreducibles $\rho$ and valid level-$D$ configurations $\tau$. For every valid configuration $\tau$ of level $m \geq D$,*

$$d_{\mathrm{TV}}(\mu_\tau, U_G) \leq \min\left\{1, \tfrac{1}{2}\sqrt{|G|-1}\,\lambda^{4^{m-D}}\right\}.$$

*For the associated conditioned law $D_T$, add $p$ from Lemma 2.5 to this bound.*

*Proof.* Every sub-block of a valid configuration is valid: were a sub-block target $e$, each ancestor commutator $[e, \cdot] = e$ would vanish, forcing the root target to

be $e$. Hence each level-$D$ sub-block has leaf subgroup $G$ and operator norm $\le \lambda$ (case (2)), and submultiplicativity gives $\|M_\tau(\rho)\| \le \lambda^{4^{m-D}}$. With $\|M\|_{\mathrm{F}}^2 \le d_\rho \|M\|^2$ and $\sum_\rho d_\rho^2 = |G|$, Diaconis–Shahshahani gives $4d_{\mathrm{TV}}(\mu_\tau, U_G)^2 \le (|G|-1)\lambda^{2\cdot 4^{m-D}}$; the $D_T$ bound is Lemma 2.5. For $G = S_5$ the prefactor $\frac{1}{2}\sqrt{119} \approx 5.45$ recovers the bound of the companion paper [63]. Section 7 controls the gap as the group varies and derives uniform family rates. □

If $G$ is solvable of derived length $d$, no configuration of level at least $d$ is valid, by Lemma 3.4; this is why $G$ is assumed non-solvable throughout.

*Remark* 3.9 (Scope: all labellings). The level is uniform over all labellings of the leaf positions by elements of $C$; in Lemma 3.5 this is what allows the two sub-blocks of a node to vary independently. A branching program for a specific function prescribes particular labellings at each level. For such a prescribed family the value $D$ of Theorem 3.7 remains an upper bound for the first level of strict contraction, by the proof of part (2), but it need not be attained: over the transpositions of $S_5$, where $D = 3$, prescribed families in which validity persists at every level realise exactly the thresholds 2 and 3. This, and the persistence of validity under conjugated doubling, are treated in the extended version [61, Section 5.1].

*Remark* 3.10 (One permutation representation is not enough). Let $G$ act transitively on a set $\Omega$ and let $V = \{f : \Omega \to \mathbb{C} : \sum_\omega f(\omega) = 0\}$. The space $V^H$ has dimension one less than the number of $H$-orbits, so by Lemma 3.1 a configuration $\tau$ has $\|M_\tau(V)\| = 1$ if and only if $H(\tau)$ is intransitive. A proper transitive leaf subgroup is therefore invisible to $V$, although by Theorem 3.2 it has a fixed vector in some other irreducible representation. For example, let $S_5$ act on the six points of the projective line over $\mathbb{F}_5$ and let $C$ be the set of all involutions. The double transpositions generate $A_5$, which is perfect and transitive on the six points; by Lemma 3.5 there are valid configurations with leaf subgroup $A_5$ at every level, and their operators have norm less than one on $V$ and norm one on the sign character. For transposition leaves in the natural action of $S_5$ this does not happen, since a proper subgroup generated by transpositions is a Young subgroup and is intransitive.

3.3. **Quotients.** The dichotomy is monotone under quotients: an obstruction visible in a quotient is already an obstruction in $G$.

**Proposition 3.11** (Quotients)**.** *Let $N \trianglelefteq G$ and $\bar{C} = \pi(C) \subseteq G/N$ its image under $\pi : G \to G/N$ (again conjugation-closed). If $(G, C)$ is in case* (2) *of Theorem 3.7, then every proper $\bar{C}$-generated subgroup of $G/N$ is solvable. In particular, if $G/N$ is non-solvable, then $(G/N, \bar{C})$ is also in case* (2)*, and the level cannot rise:*

$$D(G/N, \bar{C}) \le D(G, C).$$

*Equivalently, an obstruction in a quotient lifts to one in $G$.*

*Proof.* Let $\bar{K} < G/N$ be proper and $\bar{C}$-generated, and set $L = \langle C \cap \pi^{-1}(\bar{K}) \rangle$. Then $L$ is $C$-generated, and since $\pi\big(C \cap \pi^{-1}(\bar{K})\big) = \bar{C} \cap \bar{K}$ we have $\pi(L) = \langle \bar{C} \cap \bar{K} \rangle = \bar{K}$. Hence $L < G$ is proper and $\mathrm{dl}(L) \ge \mathrm{dl}(\pi(L)) = \mathrm{dl}(\bar{K})$. So a non-solvable $\bar{K}$ forces a non-solvable $L$ (the obstruction lifts); and if every proper $C$-generated subgroup

of $G$ is solvable, so is $L$, whence $\bar{K}$ is solvable with $\mathrm{dl}(\bar{K}) \le \mathrm{dl}(L) \le D(G, C)$. Maximising over $\bar{K}$ gives the level bound when $G/N$ is non-solvable. □

3.4. **Stability and comparison with Cayley walks.** The fixed-vector criterion has a quantitative form: a vector that loses little norm is close to the fixed space of the leaf subgroup. The estimate applies to leaves of arbitrary order.

**Proposition 3.12** (Stability / quantitative gap)**.** *Let $M_j = (I + \rho(s_j))/2$, $T = M_N \cdots M_1$, and $H = \langle s_1, \dots, s_N \rangle \neq \{e\}$, where $\rho$ is unitary on $V$ and the $s_j$ may have arbitrary orders. Let $L$ be the diameter of $H$ in $\{s_1, \dots, s_N\}^{\pm 1}$. For a unit vector $v$, put $E = 1 - \|Tv\|^2$. Then*

$$\|(I - M_j)v\| \le \sqrt{NE}, \qquad \mathrm{dist}(v, V^H) \le 2L\sqrt{NE}.$$

*In particular, if $V^H = 0$, then*

$$\|T\|^2 \le 1 - \frac{1}{4NL^2}, \qquad \|T\| \le 1 - \frac{1}{8NL^2}. \tag{3.1}$$

*Proof.* For a unitary $U$ and $M = (I + U)/2$, the parallelogram identity gives $\|Mx\|^2 + \|(I - M)x\|^2 = \|x\|^2$, and $\|I - M\| \le 1$. Set $w_0 = v$, $w_j = M_j w_{j-1}$ and $e_j = w_{j-1} - w_j$. Thus $\sum_j \|e_j\|^2 = E$ and

$$\|(I - M_j)v\| \le \|e_j\| + \|v - w_{j-1}\| \le \sum_{i \le j} \|e_i\| \le \sqrt{NE}.$$

Consequently $\|(I - \rho(s_j))v\| \le 2\sqrt{NE}$; the same holds for $s_j^{-1}$. Telescoping a word of length at most $L$ gives $\|(I - \rho(h))v\| \le 2L\sqrt{NE}$ for every $h \in H$. Averaging over $H$ proves the distance bound. If $V^H = 0$, this yields $1 \le 2L\sqrt{NE}$ for every unit $v$, proving the first inequality in (3.1); use $\sqrt{1 - x} \le 1 - x/2$ for the second. □

*Remark* 3.13. For generating leaves in a non-trivial irreducible, Proposition 3.12 gives an explicit gap in terms of their Cayley diameter. With $N = 4^d$ at level $d$, this estimate yields the family rates in Section 7.

**Proposition 3.14** (Comparison with the leaf Cayley walk)**.** *Let $s_1, \dots, s_N$ be involutions in a finite group $G$, listed with multiplicity and $N \ge 1$. On $\ell_0^2(G)$ let $R$ be the regular representation and put*

$$P_i = \tfrac{1}{2}(I + R(s_i)), \qquad T = P_N \cdots P_1, \qquad A = \frac{1}{N} \sum_{i=1}^{N} P_i, \qquad \delta = 1 - \|A\|.$$

*Then*

$$\frac{2\delta}{N+1} \;\le\; 1 - \|T\|^2 \;\le\; \min\{1, 4N\delta\}.$$

*Here $A$ is the positive semidefinite transition operator of the lazy Cayley walk with the indicated multiplicities. The assertions include $\delta = 0$, which occurs exactly when the leaves generate a proper subgroup.*

*Proof.* For a unit vector $v$ write $w_0 = v$, $w_i = P_i w_{i-1}$ and $e_i = (I - P_i)w_{i-1}$. Orthogonality gives $\sum_i \|e_i\|^2 = 1 - \|Tv\|^2$. Also

$$\|(I - P_i)v\| \le \sum_{j \le i} \|e_j\|, \qquad \sum_i \|(I - P_i)v\|^2 \le \frac{N(N+1)}{2} \sum_j \|e_j\|^2.$$

Since $I - A \ge \delta I$, the left sum is at least $N\delta$. Choose $v$ attaining $\|T\|$ to obtain the lower bound. For the upper bound choose a unit eigenvector $v$ of $I - A$ with eigenvalue $\delta$. Gao's projection union bound, in the form [3, Lemma 2], gives

$$\|Tv\|^2 \ge 1 - 4 \sum_i \|(I - P_i)v\|^2 = 1 - 4N\delta.$$

The additional bound by 1 follows from $\|T\|^2 \ge 0$. □

**Corollary 3.15** (Bounded-depth uniformity)**.** *For a family of involutive independent-read blocks with uniformly bounded expanded length $N$, their operator norms on $\ell_0^2(G)$ are uniformly bounded below 1 if and only if the associated lazy Cayley walks have uniformly positive spectral gaps. In a level-$k$ block $N = 4^k$; the walk uses the expanded occurrences, equivalently the $2^k$ leaf positions with their multiplicities. Thus the assertion applies to uniformly bounded depths, and asserts no such equivalence with uniform constants when depths are unbounded.*

*Proof.* Both implications follow from Proposition 3.14; the regular representation contains every non-trivial irreducible, so its norm on $\ell_0^2(G)$ is the maximum of their norms. □

## 4. Exact levels in families

By Theorem 3.7, determining the level of a pair $(G, C)$ is a question about subgroups: one has to find the proper $C$-generated subgroups and their derived lengths. This section does so for symmetric and alternating groups, for the minimal simple groups, for transvections in $\mathrm{SL}_n(q)$ and for congruence quotients of $\mathrm{GL}_2$. The involution classes in rank one are treated in Section 5, and Table 1 collects the resulting levels.

### 4.1. Symmetric and alternating groups.

**Corollary 4.1** (Symmetric groups: obstruction at every depth)**.** *For $G = S_n$, $n \ge 5$, with $C$ the class of transpositions, a uniform strict-contraction level exists exactly when $n = 5$, and its exact value is 3. For every $n \ge 6$ and every depth $m \ge 0$, there is a valid configuration $\tau$ with all leaves in the point stabilizer $S_{n-1} < S_n$ whose operator satisfies*

$$\|M_\tau(\rho_{\mathrm{std}})\| = 1,$$

*where $\rho_{\mathrm{std}}$ is the standard representation of $S_n$. Consequently no depth gives strict contraction uniformly over all valid configurations for any $n \ge 6$.*

*Proof.* A subgroup generated by transpositions is a Young subgroup $S_{a_1} \times \cdots \times S_{a_r}$, corresponding to the connected components of the transposition graph. For $n = 5$,

every proper such subgroup is solvable of derived length at most 3, and the point stabilizer $S_4$ attains 3; hence Theorem 3.7 gives the exact level.

For $n \geq 6$, let $H = S_{n-1}$ fix the last point. This is a proper non-solvable subgroup generated by transpositions. By Lemma 3.5, the targets of configurations with leaves in the transposition class of $H$ generate $H^{(m)} \neq 1$ at every depth $m$. Thus at least one such configuration is valid at each depth. Every leaf fixes

$$v = (1, \ldots, 1, -(n-1)) \neq 0, \qquad v \in \{x \in \mathbb{C}^n : \textstyle\sum_i x_i = 0\},$$

so every leaf projection, and therefore the entire operator, fixes $v$. As the operator is a product of contractions, its standard-representation norm is exactly 1. This gives the claimed obstruction at every depth. □

Exact shuffles by freely chosen lazy transpositions exist for every $S_n$ [2], with suitable probabilities. Corollary 4.1 identifies when fair independent reads guarantee strict contraction over all valid commutator trees.

**Corollary 4.2** (The alternating group $A_5$)**.** *For $G = A_5$ with $C$ the class of involutions, the exact level is* 2*.*

*Proof.* The proper $C$-generated subgroups are $C_2$, $V_4$, $S_3$ and $D_{10}$, all solvable, with $\max \mathrm{dl} = \mathrm{dl}(S_3) = \mathrm{dl}(D_{10}) = 2$. The subgroup $A_4$ is not $C$-generated, since its involutions generate only $V_4$. Hence $D = 2$ by Theorem 3.7(2). □

4.2. **Minimal simple groups.**

**Corollary 4.3** (Minimal simple groups)**.** *Let $G$ be a finite non-abelian simple group. The following are equivalent:*
*(1) $G$ is minimal simple: every proper subgroup is solvable.*
*(2) Every non-trivial conjugacy class has a uniform strict-contraction level.*
*(3) Every non-trivial conjugacy class has uniform strict contraction at level* 5*.*
*When these conditions hold, the exact level from Theorem 3.7 satisfies $D(G, C) \leq 5$ for every non-trivial conjugacy class $C$. The universal bound* 5 *is attained by the involution class of* $\mathrm{PSL}_3(3)$ *(Proposition 4.4).*

*Proof.* Every non-trivial conjugacy class $C$ generates $G$, since $\langle C \rangle$ is non-trivial and normal. If $G$ is minimal simple, all proper $C$-generated subgroups are solvable and have derived length at most 5 [23, Remark 2.3]. Theorem 3.7 gives the exact-level formula and (1)⇒(3). The implication (3)⇒(2) is immediate.

For (2)⇒(1), suppose that $H < G$ is non-solvable. Choose $x \in H \setminus O_\infty(H)$, where $O_\infty(H)$ is the largest soluble normal subgroup of $H$, and set $C = x^G$ and $L = \langle C \cap H \rangle$. Conjugation by $H$ preserves $C \cap H$, so $L \trianglelefteq H$. Since $x \in L$, solvability of $L$ would contradict $x \notin O_\infty(H)$. Moreover $L \leq H < G$ and $C \cap L = C \cap H$, so $L$ is a proper non-solvable $C$-generated subgroup. Theorem 3.7(1) gives an obstruction at every depth, contradicting (2). □

Thus contraction for every non-trivial class singles out precisely the groups in Thompson's classification [59]: $\mathrm{PSL}_2(2^p)$ ($p$ prime), $\mathrm{PSL}_2(3^p)$ ($p$ odd prime), $\mathrm{PSL}_2(p)$ for primes $p > 3$ with $p \equiv \pm 2 \pmod 5$, the Suzuki groups $\mathrm{Sz}(2^p)$ ($p$ odd prime), and $\mathrm{PSL}_3(3)$.

**Proposition 4.4** (The involution class of $\mathrm{PSL}_3(3)$)**.** *For $G = \mathrm{PSL}_3(3)$ and its involution class, the exact level is* 5.

*Proof.* Here $G = \mathrm{SL}_3(3)$ has trivial centre. Its involutions have eigenvalues $1, -1, -1$, form a single conjugacy class, and generate $G$ by simplicity. The maximal-subgroup list in the ATLAS [60] consists of two parabolic classes $3^2{:}\,\mathrm{GL}_2(3)$, one class of $S_4$, and 13:3.

Write a point parabolic as $P = V \rtimes L$, where $V \cong \mathbb{F}_3^2$ and $L \cong \mathrm{GL}_2(3)$. In the Levi factor,

$$L' = \mathrm{SL}_2(3), \qquad \mathrm{SL}_2(3)' = Q_8, \qquad Q_8' = \langle z \rangle, \qquad z = -I.$$

The element $z$ acts as negation on $V$, so $[V, \langle z \rangle] = V$. Consequently the derived series of $P$ is

$$P > V \rtimes \mathrm{SL}_2(3) > V \rtimes Q_8 > V \rtimes \langle z \rangle > V > 1,$$

and $\mathrm{dl}(P) = 5$.

The group $L$ is generated by $s = \mathrm{diag}(-1, 1)$ and the elementary transvections $t = I + E_{12}, I + E_{21}$. Since $sts = t^{-1}$ for either transvection, both $s$ and $st$ are involutions, so $L$ is involution-generated. For every $v \in V$, the product $(vzv^{-1})z$ is the translation $2v$. Thus the involutions of $P$ also generate $V$, and hence $P$. The line parabolic has the same properties by inverse transpose.

The remaining maximals are $S_4$, of derived length 3, and the odd-order group 13:3. Every proper involution-generated subgroup therefore has derived length at most 5, and a parabolic attains 5. Apply Theorem 3.7. □

*Remark* 4.5 ($\mathrm{PSL}_2(7)$). The group $\mathrm{GL}_3(\mathbb{F}_2) \cong \mathrm{PSL}_2(7)$ is minimal simple, with maximal subgroups two classes of $S_4$ and one of 7:3. For the involution class the proper $C$-generated subgroups have maximal derived length $\mathrm{dl}(S_4) = 3$, the group 7:3 having odd order, so $D = 3$ by Theorem 3.7(2).

4.3. **Transvections.** For $G = \mathrm{GL}_n(\mathbb{F}_q)$ or $\mathrm{SL}_n(\mathbb{F}_q)$ the natural candidates for obstructions are subspace stabilisers. If $P$ is a proper parabolic subgroup and the image of $\langle P \cap C \rangle$ in a Levi factor of $P$ is non-solvable, then $\langle P \cap C \rangle$ is a proper non-solvable $C$-generated subgroup, and Theorem 3.7(1) applies. For transvections this settles all but two cases.

**Proposition 4.6** (Transvection leaves)**.** *Let $n \geq 3$, let $G = \mathrm{SL}_n(q)$, and let $\mathcal{T}$ be its class of transvections. The exact level is* 3 *for $(n, q) = (3, 2)$ and* 4 *for $(n, q) = (3, 3)$. In every other case there is a permanent obstruction.*

*Proof.* Transvections generate $\mathrm{SL}_n(q)$. For $q \geq 4$ the upper-left $\mathrm{SL}_2(q)$ is a proper non-solvable $\mathcal{T}$-generated subgroup. For $q \in \{2, 3\}$ and $n \geq 4$, use the upper-left $\mathrm{SL}_3(q)$ instead. Apply Theorem 3.7(1). In $\mathrm{SL}_3(2)$ transvections are precisely the involutions, and Remark 4.5 gives level 3.

For $\mathrm{SL}_3(3) = \mathrm{PSL}_3(3)$ use the maximal-subgroup list in Proposition 4.4. In a point parabolic $P = 3^2{:}\,\mathrm{GL}_2(3)$, a transvection acts trivially on the invariant line, so its Levi image lies in $\mathrm{SL}_2(3)$. The radical $V \cong \mathbb{F}_3^2$ and the elementary matrices

in this Levi factor are transvections; hence $\langle P \cap \mathcal{T} \rangle = J = V \rtimes \mathrm{SL}_2(3)$. Since $\mathrm{SL}_2(3)' = Q_8$, $Q_8' = \langle -I \rangle$, and $-I$ acts as negation on $V$, the derived series is

$$J \;>\; V \rtimes Q_8 \;>\; V \rtimes \langle -I \rangle \;>\; V \;>\; 1,$$

with orders $216, 72, 18, 9, 1$. The dual parabolic has the same property. The other maximals, $S_4$ and 13:3, have derived lengths 3 and 2. Every proper $\mathcal{T}$-generated subgroup therefore has derived length at most 4, and $J$ attains 4. Apply Theorem 3.7(2). □

### 4.4. **Abelian and congruence quotients.**

*Remark* 4.7 (One-dimensional representations). Let $\chi$ be a one-dimensional representation of $G$ and let $w(\tau) = (g_1, \ldots, g_N)$. By Lemma 2.1,

$$M_\tau(\chi) = \prod_{j=1}^{N} \tfrac{1}{2}\big(1 + \chi(g_j)\big), \qquad |M_\tau(\chi)| = \prod_{j=1}^{N} |\cos(\theta_j/2)|,$$

where $\chi(g_j) = e^{i\theta_j}$. This has modulus one exactly when $\chi$ is trivial on every leaf, in accordance with Lemma 3.1. In particular, if some leaf lies outside the derived subgroup $G'$, the output law is not supported on a single coset of $G'$. For $G = \mathrm{GL}_n(\mathbb{F}_q)$ with $q > 3$, involutions have determinant $\pm 1$ and cannot generate $G$, so a generating leaf set must contain elements whose determinants generate $\mathbb{F}_q^\times$.

Let $p$ be a prime and $r \geq 2$. Reduction modulo $p$ is a surjection $\mathrm{GL}_2(\mathbb{Z}/p^r) \to \mathrm{GL}_2(\mathbb{F}_p)$ whose kernel $K = 1 + pM_2(\mathbb{Z}/p^r)$ is a $p$-group, so $\mathrm{GL}_2(\mathbb{Z}/p^r)$ is non-solvable exactly when $p \geq 5$. For conjugacy classes, Serre's lifting lemma reduces the dichotomy to the residue field.

**Proposition 4.8** (Congruence lifting with generating determinants)**.** *Let $p \geq 5$ be prime, $r \geq 2$, $G = \mathrm{GL}_2(\mathbb{Z}/p^r)$, and $K = \ker(G \to \mathrm{GL}_2(\mathbb{F}_p))$. Let $C$ be conjugation-closed, generate $G$, and have the property that* $\det c$ *generates $(\mathbb{Z}/p^r)^\times$ for every $c \in C$. Then $(G, C)$ has a uniform contraction level if and only if $(\mathrm{GL}_2(\mathbb{F}_p), \bar{C})$ does. In this case their exact levels satisfy*

$$\bar{D} \leq D \leq \bar{D} + \mathrm{dl}(K).$$

*In particular, $\bar{D} \leq D \leq \bar{D} + 1$ for $r = 2$. The determinant hypothesis holds for every single conjugacy class that generates $G$.*

*Proof.* Every non-trivial $C$-generated subgroup $H$ has full determinant image. If its reduction is $\mathrm{GL}_2(\mathbb{F}_p)$, Serre's lifting lemma [48, Lemma 3.15], applied to the closed preimage of $H$ in $\mathrm{GL}_2(\mathbb{Z}_p)$, implies that $H$ contains $\mathrm{SL}_2(\mathbb{Z}/p^r)$. The full determinant image then gives $H = G$. Consequently every proper $C$-generated subgroup has proper reduction. Since $K$ is solvable, a non-solvable obstruction above yields one below; conversely, Proposition 3.11 lifts every obstruction below. In the unobstructed case,

$$\mathrm{dl}(H) \leq \mathrm{dl}(\bar{H}) + \mathrm{dl}(H \cap K) \leq \bar{D} + \mathrm{dl}(K)$$

for every proper $C$-generated $H$. The opposite inequality $\bar{D} \leq D$ follows from Proposition 3.11. For $r = 2$, $K \cong M_2(\mathbb{F}_p)^+$ is abelian. Finally, all elements of a

single conjugacy class have the same determinant, which must generate the unit group if the class generates $G$. □

*Remark* 4.9 (A congruence obstruction for a union of classes). In $G = \mathrm{GL}_2(\mathbb{Z}/25)$ put $c = \mathrm{diag}(7, 1)$ and $C = c^G \cup \{6I\}$. Then $H = \langle c^G \rangle = \det^{-1}(\langle 7 \rangle)$ is a proper non-solvable $C$-generated subgroup of index 5. Indeed, conjugating $c$ by elementary unipotents and multiplying by $c^{-1}$ gives all upper and lower unipotents: the respective coefficients $1 - 7 = -6$ and $1 - 7^{-1} = 8$ are units modulo 25. They generate $\mathrm{SL}_2(\mathbb{Z}/25)$, proving the assertion about $H$. The determinants 7 and $\det(6I) = 11$ have orders 4 and 5, so $C$ generates $G$. Its reduction $\bar{C} = \mathrm{diag}(2, 1)^{\mathrm{GL}_2(5)} \cup \{I\}$ is unobstructed. Every non-trivial $\bar{C}$-generated subgroup has full determinant image; every non-solvable subgroup of $\mathrm{GL}_2(5)$ contains $\mathrm{SL}_2(5)$ by Dickson's classification [48, Theorem 3.13], so these two properties force it to be the whole group.

*Remark* 4.10 (Questions). Two questions about matrix groups remain open. For $n \geq 3$, which generating conjugacy classes of $\mathrm{GL}_n(\mathbb{F}_q)$ with irreducible characteristic polynomial have only solvable proper class-generated subgroups? Such elements stabilise no proper non-zero subspace, so the question concerns the non-parabolic maximal subgroups, in contrast with Proposition 4.6. Secondly, for a generating conjugacy class of $\mathrm{GL}_2(\mathbb{Z}/p^2)$, $p \geq 5$, with unobstructed reduction, when is the level in Proposition 4.8 equal to $\bar{D}$ and when to $\bar{D} + 1$?

## 5. Involution classes in rank one

Here the dichotomy gives an explicit classification for an infinite family. Throughout this section $q = p^f \geq 5$ is odd. In $\mathrm{PGL}_2(q)$ the determinant modulo squares defines a character with kernel $\mathrm{PSL}_2(q)$. Call an involution *interior* or *exterior* according as its determinant is a square or a non-square. These are the two involution classes of $\mathrm{PGL}_2(q)$. The subgroup classification used below is Dickson's, in the form recorded by Giudici [26, Theorems 2.2 and 3.5]; our conclusions concern the commutator-tree levels for these specified leaf sets.

**Lemma 5.1** (Classes and subfields)**.** *The interior involutions generate* $\mathrm{PSL}_2(q)$*, and the exterior involutions generate* $\mathrm{PGL}_2(q)$*. Every* $A_5$ *subgroup of* $\mathrm{PGL}_2(q)$ *is interior. For a proper subfield* $\mathbb{F}_{q_0} \subset \mathbb{F}_q$*, exterior involutions of* $\mathrm{PGL}_2(q_0)$ *remain exterior precisely when* $[\mathbb{F}_q : \mathbb{F}_{q_0}]$ *is odd.*

*Proof.* The involutions of the simple group $\mathrm{PSL}_2(q)$ form a non-trivial normal subset, so generate it. The subgroup generated by the exterior class is normal in $\mathrm{PGL}_2(q)$. Its intersection with the simple socle is either the socle or trivial; in the latter case its commutator with the socle is trivial, contradicting the trivial centralizer of the socle in $\mathrm{PGL}_2(q)$. Hence it contains the socle and an exterior element, and is the whole group. The determinant character is trivial on the perfect group $A_5$. Finally, if $m = [\mathbb{F}_q : \mathbb{F}_{q_0}]$, the parity of $(q - 1)/(q_0 - 1) = 1 + q_0 + \cdots + q_0^{m-1}$ is the parity of $m$. The square-character restriction therefore has the stated parity. □

**Theorem 5.2** (Rank-one classification)**.** *For odd* $q \geq 5$ *the following hold.*

(1) *In* $G = \mathrm{PGL}_2(q)$*, the set of* all *involutions has a permanent obstruction.*
(2) *In* $G = \mathrm{PGL}_2(q)$*, the exterior class has a permanent obstruction if and only if there is a proper subfield* $\mathbb{F}_{q_0}$ *with* $q_0 \geq 5$ *and* $[\mathbb{F}_q : \mathbb{F}_{q_0}]$ *odd. Otherwise its exact level is*

$$D = \begin{cases} 3, & q \equiv \pm 3 \pmod 8, \\ 2, & q \equiv \pm 1 \pmod 8. \end{cases}$$

(3) *In* $G = \mathrm{PSL}_2(q)$*, the involution class has a permanent obstruction if and only if* $G$ *contains a* proper *subgroup isomorphic to* $A_5$*, or* $\mathbb{F}_q$ *has a proper subfield of size* $q_0 \geq 5$*. Otherwise its exact level is*

$$D = \begin{cases} 3, & q \equiv \pm 1 \pmod 8, \\ 2, & q \equiv \pm 3 \pmod 8. \end{cases}$$

*In particular* $\mathrm{PSL}_2(5) = A_5$ *has level* 2*. The* $A_5$ *obstruction in this part occurs exactly when* $q > 5$ *and either* $p = 5$ *or* $q \equiv \pm 1 \pmod 5$.

*Proof.* For (1), the interior leaves generate the proper non-solvable subgroup $\mathrm{PSL}_2(q)$, so Theorem 3.7 applies.

For (2), a proper exterior-generated subgroup lies in a maximal subgroup other than $\mathrm{PSL}_2(q)$. Giudici's Theorem 3.5 lists these maximals: an affine group $\mathbb{F}_q^+ \rtimes \mathbb{F}_q^\times$, torus normalizers (dihedral groups), $S_4$ when $q = p \equiv \pm 3 \pmod 8$, and $\mathrm{PGL}_2(q_0)$ when $q = q_0^r$ with $r$ an odd prime. The affine group is metabelian, including over extension fields; its involution-generated subgroups need not be ordinary dihedral groups. Torus normalizers are also metabelian. The only non-solvable possibility outside the socle is therefore a proper subfield group with $q_0 \geq 5$ and odd degree. Such a group is indeed exterior-generated by Lemma 5.1. Any odd proper degree has an odd prime divisor, so the existence criterion agrees with this maximal-subgroup list.

In its absence, the possible subfield maximal with $q_0 = 3$ is $\mathrm{PGL}_2(3) = S_4$, occurring for $q = 3^r$, $r$ an odd prime. Together with the prime-field $S_4$ case, these give an exterior-generated $S_4$ exactly in the unobstructed cases $q \equiv \pm 3 \pmod 8$. (Indeed, this congruence forces $f$ odd; without a proper subfield of size at least 5 and odd degree, $f = 1$ when $p \geq 5$, and $f$ is an odd prime when $p = 3$.) Its determinant character is the sign character, so its transpositions are exterior and its derived length 3 is attained. In the remaining cases all relevant maximals are metabelian. The upper bound 2 is attained: for $q \geq 7$, take a split torus generator $r : x \mapsto ax$ and an exterior involution $t : x \mapsto b/x$ ($-b$ non-square). The two exterior involutions $t, r^2t$ generate $\langle r^2, t\rangle$, a proper non-abelian dihedral group whose rotation subgroup has order $(q-1)/2 \geq 3$. The case $q = 5$ has the exterior $S_4$ and was already covered. Apply Theorem 3.7 for exactness.

For (3), Dickson's list says that any proper non-solvable subgroup lies in an $A_5$ or a non-solvable subfield subgroup. A proper $A_5$ is generated by its involutions. Every proper subfield of size $q_0 \geq 5$ supplies a proper involution-generated $\mathrm{PSL}_2(q_0)$, irrespective of the degree; even degree can also supply $\mathrm{PGL}_2(q_0)$. Conversely, the absence of these two obstructions leaves affine and dihedral subgroups of derived length at most 2, and $A_4$ or $S_4$. The involutions in $A_4$ generate only $V_4$. An $S_4$ occurs

precisely for $q \equiv \pm 1 \pmod 8$ and is generated by involutions, giving level 3 when unobstructed. Otherwise a non-abelian dihedral torus normalizer gives level 2: at least one of $(q-1)/2, (q+1)/2$ is at least 3, and its dihedral normalizer is proper. This argument uses existence, not maximality, of these dihedral groups at small $q$. At $q = 5$ the $A_5$ is the ambient group and is not an obstruction, in agreement with Corollary 4.2. The stated $A_5$ embedding condition is the corresponding part of Dickson's classification. □

**Corollary 5.3** (Prime fields)**.** *For every prime $p \geq 5$, the exterior involution class of* $\mathrm{PGL}_2(p)$ *mixes at exact level* 3 *for* $p \equiv \pm 3 \pmod 8$ *and at exact level* 2 *for* $p \equiv \pm 1 \pmod 8$.

Thus $q = 5, 7, 11$ give exterior levels $3, 2, 3$. At $q = 11$ the involutions of the $\mathrm{PSL}_2(11)$ ambient are instead obstructed by a proper $A_5$. The extension degree matters: $q = 25$ has exterior level 2, whereas $q = 125$ contains the exterior-generated non-solvable $\mathrm{PGL}_2(5)$ and is obstructed. Section 7 gives mixing rates as $q$ grows.

## 6. The natural octahedral pair: a sharp uniform angle

We construct a pair of octahedral subgroups whose fixed-space angle is controlled by the Ramanujan bound for a quaternionic Cayley walk, and show that it is the only pair of its kind up to conjugacy. A compression identity connects the two operators.

6.1. **Fixed-space angles and double cosets.** Let $H$ be a subgroup of a finite group $G$ and let $\rho$ be a unitary representation of $G$ on $V$. Put

$$e_H = \frac{1}{|H|} \sum_{h \in H} h \in \mathbb{C}[G], \qquad P_H = P_H^\rho = \rho(e_H) = \frac{1}{|H|} \sum_{h \in H} \rho(h).$$

When $\rho$ is fixed we omit the superscript; we retain it when taking maxima over representations.

**Lemma 6.1** (Double cosets and angles between fixed spaces)**.** *The operator $P_H$ is the orthogonal projection onto $V^H$. For $g \in G$ and $K = gHg^{-1}$,*

$$P_K = \rho(g) P_H \rho(g)^{-1}, \qquad \|P_H \rho(g) P_H\| = \|P_H P_K\|.$$

*The element $e_H g e_H$ depends only on the double coset $HgH$.*

*Proof.* The average $P_H$ is self-adjoint and idempotent. Its image is $H$-fixed by reindexing the sum, and it acts as the identity on every $H$-fixed vector. Conjugating the average over $H$ gives the formula for $P_K$, so $P_H \rho(g) P_H = P_H P_K \rho(g)$, and right multiplication by the unitary $\rho(g)$ preserves the norm. Finally $e_H h_1 g h_2 e_H = e_H g e_H$ for $h_1, h_2 \in H$. □

Thus $\|P_H \rho(g) P_H\|$, the norm in $\rho$ of the element of the Hecke algebra $e_H \mathbb{C}[G] e_H$ attached to the double coset $HgH$, is the cosine of the smallest principal angle between $V^H$ and $V^K$, with cosine 0 when either space is zero. If $\langle H, K \rangle = G$ and $\rho$ is non-trivial and irreducible, then $V^H \cap V^K = V^G = 0$, and this is the Friedrichs angle, which governs the alternation of the two projections [5, 38].

6.2. **The pair and its four neighbours.** For the rest of this section $q$ is a prime with $q \equiv \pm 3 \pmod 8$, $q \neq 3$, and $G_q = \mathrm{PGL}_2(\mathbb{F}_q)$. Let $\mathbb{H} = (-1,-1)_{\mathbb{Q}}$, with $i^2 = j^2 = k^2 = ijk = -1$, and take projective classes in $\mathbb{H}^\times/\mathbb{Q}^\times$:

$$A_0 = \langle i, 1+i, 1+i+j+k\rangle, \qquad h_0 = i+j+k, \qquad S_0 = \{\pm i \pm j \pm k\}/\{\pm 1\}.$$

The four elements of $S_0$ have norm 3 and are projective involutions. For any odd prime $q$, the quaternion algebra over $\mathbb{F}_q$ is split. Indeed, the two sets of squares and of $-1$ minus squares have more than $q$ elements in total, so $x^2 + y^2 = -1$ has a solution; this gives an explicit splitting into $M_2(\mathbb{F}_q)$ [19, Proposition 2.5.2]. Reducing the indicated projective elements defines $A, h, S$ in $G_q$. Their norms $1, 2, 4, 3$ are invertible for our primes. Changing the splitting conjugates the entire construction, leaving the norms below unchanged.

**Lemma 6.2** (Octahedral reduction)**.** *One has $A \cong S_4$, with exterior transpositions, and*

$$B = hAh^{-1}, \quad A \cap B \cong S_3, \quad h^2 = 1, \quad \langle A, B\rangle = G_q, \quad AhA/A = \{sA : s \in S\}.$$

*The last set has four distinct cosets, and $S$ is stable under conjugation by $A$.*

*Proof.* On the trace-zero subspace with basis $(i, j, k)$, conjugation by the three generators of $A_0$ and by $h_0$ gives, respectively,

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad H = \frac{1}{3}\begin{pmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{pmatrix}.$$

The first three generate the 24 determinant-one signed permutation matrices. Their action on the four body diagonals identifies them with $S_4$; they remain distinct modulo every odd prime. The adjoint action is faithful projectively, also in odd characteristic. Conjugation by $\alpha \in \mathbb{H}(\mathbb{F}_q)^\times$ is $L_\alpha R_{\alpha^{-1}}$ on $\mathbb{H}(\mathbb{F}_q)$, of determinant $N(\alpha)^2 N(\alpha)^{-2} = 1$ for the norm $N$; it fixes 1 and preserves the norm form, so the adjoint action embeds $G_q$ into $\mathrm{SO}_3(\mathbb{F}_q)$. The norm of $1+i$ is 2, a nonsquare modulo $q$, so the determinant-square character is non-trivial on $A$ and hence is its sign character. In particular the transpositions of $A$ are exterior.

Here is a uniform intersection calculation. Let $K$ be the stabilizer in $A$ of the line through $(1, 1, 1)$: its matrices are the positive even permutation matrices and the negative odd permutation matrices. Thus $K \cong S_3$, and every element of $K$ commutes with $H$. Let $V$ be the four diagonal sign matrices of determinant one. Then $A = VK$, and $V \cap K = 1$. The three non-identity elements of $V$ are conjugate under $K$. For $d = \mathrm{diag}(1, -1, -1)$,

$$HdH = \frac{1}{9}\begin{pmatrix} -7 & -4 & -4 \\ -4 & -1 & 8 \\ -4 & 8 & -1 \end{pmatrix}.$$

Its second row has three nonzero entries modulo every prime $q \geq 5$, so $HdH \notin A$. Since $H$ commutes with $K$, the same conclusion holds for the other two non-identity cosets of $K$ in $A$. Therefore $A \cap HAH = K$, in characteristic zero and modulo every

allowed prime. This also proves $h \notin A$ and $B \neq A$. The exterior $S_4$ is maximal in $G_q$ for these primes by [26, Theorem 3.5]; hence $\langle A, B\rangle = G_q$.

Finally $H^2 = I$, and $A$ acts transitively by conjugation on the four body diagonals $S$, with stabilizer $K$. The map $A/K \longrightarrow AhA/A$, $aK \mapsto ahA$, is bijective by the intersection calculation. Since $(aha^{-1})A = ahA$, its images are exactly the four cosets represented by $S$. □

**Proposition 6.3** (The intersection-six stratum)**.** *Let $A_1 \neq A_2$ be exterior octahedral subgroups of $G_q$. Then $|A_1 \cap A_2| \leq 6$. If $|A_1 \cap A_2| = 6$, the ordered pair $(A_1, A_2)$ is simultaneously conjugate in $G_q$ to $(A, hAh^{-1})$, and for fixed $A_1$ there are exactly four such $A_2$.*

*Proof.* The exterior octahedral subgroups form one conjugacy class (Dickson's classification of the subgroups of $\mathrm{PSL}_2(q)$ and $\mathrm{PGL}_2(q)$ [34, 40]; for the statement in this form see [26, proof of Proposition 3.4]), so we may take $A_1 = A$ and $A_2 = gAg^{-1}$ with $g \notin A$, in the orthogonal model of Lemma 6.2.

*Normalisers.* The three non-identity elements of $V$ have the coordinate axes as fixed lines, so an element normalising $V$ permutes these axes. Orthogonality forces its nonzero entries to be $\pm 1$, and determinant one leaves the 24 elements of $A$. Thus $N_{G_q}(V) = A$; since $V$ is characteristic in $A$, also $N_{G_q}(A) = A$.

*Orders* 8 *and* 12*.* Put $D = A \cap A_2$. The determinant-square character restricts to the sign character of $A$ and of $A_2$, so $A \cap \mathrm{PSL}_2(q)$ and $A_2 \cap \mathrm{PSL}_2(q)$ are their $A_4$ subgroups, whose only subgroups of order 4 are their normal Klein four subgroups. If $|D| = 12$, then $D$ is the $A_4$ of both groups. If $|D| = 8$, then $D$ contains transpositions, so $D \cap \mathrm{PSL}_2(q)$ has order 4 and is the normal Klein four of both groups. In either case $A_2$ normalises $V$, so $A_2 \leq A$, a contradiction. Hence $|D| \leq 6$, and $|D| = 6$ forces $D \cong S_3$.

*Order* 6*.* The four $S_3$ subgroups of $A$ are conjugate in $A$, so replacing $g$ by $a_1 g a_2^{-1}$ with $a_1, a_2 \in A$, which does not change $AgA$, we may assume that $g$ normalises $K$. Every automorphism of $S_3$ is inner, so $N_{G_q}(K) = K\,C_{G_q}(K)$. The elements of $K$ are the matrices $\mathrm{sgn}(\pi)P_\pi$, $\pi \in S_3$, so a matrix commutes with $K$ exactly when it commutes with every permutation matrix $P_\pi$, that is, when it is constant on the two orbits of $S_3$ on index pairs: the commutant is $\{aI + bJ\}$, with $J$ the all-ones matrix. For $M = aI + bJ$, orthogonality gives $a^2 = 1$ and $b(2a + 3b) = 0$; since $q \neq 3$, $M$ acts by $a + 3b$ on $(1, 1, 1)$ and by $a$ on the complementary plane $x + y + z = 0$, so determinant one gives $a + 3b = 1$. The solutions are $I$ and $H$. Hence $C_{G_q}(K) = \{1, h\}$, $N_{G_q}(K) = K \sqcup Kh$, and $g \in Kh$ because $g \notin A$. Therefore $AgA = AhA$; writing $g = ahb$ with $a, b \in A$ gives $A_2 = a(hAh^{-1})a^{-1}$, so $(A, A_2) = a(A, hAh^{-1})a^{-1}$.

*Count.* As $N_{G_q}(A) = A$, the map $gA \mapsto gAg^{-1}$ is injective. The subgroups $A_2$ with $|A \cap A_2| = 6$ are the images of the cosets in $AhA/A$, of which there are $[A : K] = 4$ by Lemma 6.2. □

6.3. **Compression and the arithmetic spectral input.** Put $\mu = \frac{1}{4}\sum_{s\in S}\delta_s$ and let $e_A$ denote the uniform average over $A$ in $\mathbb{C}[G_q]$.

**Lemma 6.4** (The compression identity)**.** *In the group algebra,*

$$\mu e_A = e_A \mu = e_A h e_A. \tag{6.1}$$

*Consequently $\rho(\mu)$ commutes with $P_A$ and its restriction to $V_\rho^A$ is $P_A \rho(h) P_A$, whose norm equals $\|P_A P_B\|$.*

*Proof.* Conjugation invariance of $S$ gives commutation with $A$. Averaging $a h e_A$ over $a \in A$ visits each of the four cosets in Lemma 6.2 equally often, proving $e_A h e_A = \frac{1}{4} \sum_{s \in S} s e_A$. Apply $\rho$ and Lemma 6.1. □

**Proposition 6.5** (Quaternionic Ramanujan input)**.** *For $\chi_q(g)$ the determinant-square character of $G_q$ and $\epsilon_q = (3/q)$, the operator $\rho(\mu)$ has norm at most $\sqrt{3}/2$ on every irreducible $\rho \notin \{\mathbf{1}, \chi_q\}$. On those two characters its values are $1$ and $\epsilon_q$.*

*Proof.* We apply the theorem of Davidoff–Sarnak–Valette [19, Theorem 4.2.2]. Its hypotheses are distinct odd primes $p, q$ with $q > 2\sqrt{p}$, and its distinguished norm-$p$ generators are specified in Section 2.6 of that reference. For $p = 3$ they are exactly the four classes of $\pm i \pm j \pm k$, so every prime here satisfies the hypotheses. Remark 4.4.7 of the same reference derives the Ramanujan bound from Eichler's theorem on weight-two cusp forms.

The vertex set of the walk follows from the construction. The set $S$ is stable under conjugation by $A$ and contains $h$, so $\langle S \rangle$ is normalised by $\langle A, h \rangle = G_q$ and is a non-trivial normal subgroup of $\mathrm{PGL}_2(q)$. Since $\mathrm{PSL}_2(q)$ is simple with trivial centraliser in $\mathrm{PGL}_2(q)$, $\langle S \rangle$ contains $\mathrm{PSL}_2(q)$. Every $s \in S$ has determinant class 3; hence $\langle S \rangle = \mathrm{PSL}_2(q)$ when $\epsilon_q = 1$ and $\langle S \rangle = \mathrm{PGL}_2(q)$ when $\epsilon_q = -1$. These are the two vertex sets in the cited theorem, and the external input is the spectral bound alone.

If $\epsilon_q = -1$, their Cayley graph on $G_q$ is connected, bipartite and 4-regular, with all eigenvalues other than $\pm 4$ bounded in absolute value by $2\sqrt{3}$. The exceptional normalized eigenspaces are the constants and $\chi_q$. If $\epsilon_q = 1$, the theorem gives a connected nonbipartite Ramanujan graph on $\mathrm{PSL}_2(q)$. On $G_q$ the same walk is the union of two copies, so the exceptional eigenspace at 1 is spanned by the constants and $\chi_q$, and there is no eigenvalue $-1$. The regular representation contains every irreducible, giving the assertion. In both cases each $s \in S$ has determinant square class 3. □

The compression identity (6.1) identifies the Hecke operator. Left multiplication by the fixed set $S$ on the cosets $G_q/A$ is generally a different operator from the Hecke adjacency. For $q = 5$, those two unnormalized spectra are respectively $\{4, 3, 3, 3, -1\}$ and $\{4, -1, -1, -1, -1\}$. The latter graph is $K_5$.

**Theorem 6.6** (Sharp angle for the natural pair)**.** *For the pair in Lemma 6.2, set*

$$c_q = \max_{\rho \in \widehat{G_q},\, \rho \neq \mathbf{1}} \|P_A^\rho \rho(h) P_A^\rho\|.$$

*Then, for every pair $A_1, A_2$ of exterior octahedral subgroups of $G_q$ with $|A_1 \cap A_2| = 6$,*

$$\max_{\rho \neq \mathbf{1}} \|P_{A_1}^\rho P_{A_2}^\rho\| = c_q = \max_{\rho \neq \mathbf{1}} \|P_A^\rho P_B^\rho\| \leq \frac{\sqrt{3}}{2}, \qquad \lim_{q \to \infty} c_q = \frac{\sqrt{3}}{2}.$$

*Thus the constant is best possible uniformly over this family.*

*Proof.* By Proposition 6.3, $(A_1, A_2) = x(A, B)x^{-1}$ for some $x \in G_q$, and $P^{\rho}_{xAx^{-1}} = \rho(x)P^{\rho}_{A}\rho(x)^{-1}$; this gives the first equality in every representation. It remains to treat $(A, B)$. For $\rho \notin \{\mathbf{1}, \chi_q\}$, the compression identity and Proposition 6.5 give the upper bound. On $\chi_q$, $P_A = 0$ because $A$ is exterior, so this character contributes zero.

For the lower bound, define $X_q$ with vertices the left cosets $xA$, and neighbours $xsA$ for $s \in S$. This is independent of the representative $x$, since $S$ is invariant under conjugation by $A$. The four neighbours are distinct and none is $xA$, by Lemma 6.2. The relation is symmetric because $h^{-1} = h$ and $AhA = (AhA)^{-1}$. The graph is connected because $\langle A, h\rangle = G_q$, and has $|G_q|/24$ vertices. On functions on $G_q$ invariant under right multiplication by $A$, its normalized adjacency is the restriction of the *right* convolution

$$(Wf)(x) = \frac{1}{4}\sum_{s\in S} f(xs).$$

Under the finite Peter–Weyl decomposition, the multiplicity spaces of this coset representation are $V_\rho^A$ (or their duals, with identical norms), and the blocks are the compressions $P_A\rho(h)P_A$. Hence the absolute nonconstant spectral radius of $X_q$ is $4c_q$. The trivial representation contributes just the constant functions, and $\chi_q$ has no $A$-fixed vectors. Alon–Boppana [19, Theorem 1.3.1], applied to these connected loopless 4-regular graphs with vertex count tending to infinity, gives $\liminf 4c_q \geq 2\sqrt{3}$. Together with the upper bound this proves the limit. □

6.4. **From angles to tree blocks.** The minimum fixed-space angle $\theta_q = \arccos c_q$ satisfies $\theta_q \geq \pi/6$ and $\theta_q \to \pi/6$ by Theorem 6.6. This angular separation controls how the averages over $A$ and $B$ interact. The following estimate transfers it to configurations whose sub-blocks approximate those averages.

**Lemma 6.7** (Quantitative gluing)**.** *Let $P, Q$ be orthogonal projections with $\|PQ\| \leq c < 1$. Suppose $X = P + R$, $Y = Q + S$, where $PR = RP = QS = SQ = 0$ and $\|R\|, \|S\| \leq \lambda < 1$. Then*

$$\|XYX^{\dagger}Y^{\dagger}\|^2 \leq 1 - \frac{(1-\lambda)(1-c)}{8(1+\lambda)}. \tag{6.2}$$

*Proof.* Put $g = 1 - \lambda^2$. Each factor $Z$ with associated projection $F$ satisfies

$$\|w\|^2 - \|Zw\|^2 \geq g\|(I-F)w\|^2, \qquad \|w - Zw\| \leq (1+\lambda)\|(I-F)w\|.$$

Apply the four factors in their actual order to a unit $v$, and denote their norm-squared losses by $e_j \geq 0$. Their sum is $E = 1 - \|XYX^{\dagger}Y^{\dagger}v\|^2$. At each step, telescoping back to $v$ and Cauchy–Schwarz give

$$\|(I-F)v\| \leq \frac{1+\lambda}{\sqrt{g}}\sum_{i\leq j}\sqrt{e_i} \leq \frac{2(1+\lambda)}{\sqrt{g}}\sqrt{E}.$$

Both $P$ and $Q$ occur. Also $\|P + Q\| \leq 1 + c$: the map $v \mapsto (Pv, Qv)$ has squared norm bounded by that of the block matrix with identity diagonal and off-diagonal

blocks $PQ, QP$. Consequently

$$1 - c \le \|(I-P)v\|^2 + \|(I-Q)v\|^2 \le \frac{8(1+\lambda)^2}{g} E.$$

Taking the supremum over $v$ proves (6.2). □

**Corollary 6.8** (Uniform contraction for natural-pair blocks)**.** *Suppose a valid level-3 tree $\tau$ has level-2 leaf subgroups $A, B \cong S_4$, exterior with $|A \cap B| = 6$, and all leaves are exterior involutions. For every non-trivial $\rho$,*

$$\|M_\tau(\rho)\| \le \kappa, \qquad \kappa = \left(1 - \frac{183}{1736}\left(1 - \frac{\sqrt{3}}{2}\right)\right)^{1/2} < 1.$$

*For a valid level-$m$ tree, $m \ge 3$, all of whose level-3 sub-blocks have this property, replace $\lambda$ by $\kappa$ and $D$ by 3 in Proposition 3.8.*

*Proof.* On the fixed space of its leaf subgroup, each level-2 operator is the identity. On the orthogonal complement its norm is at most $\lambda = 17/200$: this is a finite rational certificate on the four non-trivial irreducibles of $S_4$, over its 432 valid ordered transposition configurations generating $S_4$. The exact certificate described in Section 2.2 constructs rational matrices with invariant Gram matrices $G_\pi$ and verifies all principal minors of $(17/200)^2 G_\pi - D^t G_\pi D$ are nonnegative. Restriction of any ambient unitary representation to $S_4$ decomposes into these irreducibles and the trivial one; hence the certificate applies in every ambient $G_q$. Both invariant subspaces are preserved, so the hypotheses of Lemma 6.7 hold. Insert $c = \sqrt{3}/2$ from Theorem 6.6. The last assertion follows by expanding into the level-3 blocks, as in Proposition 3.8. □

The sharp angle estimate and the finite $S_4$ certificate thus give the explicit contraction bound $\kappa$ for these octahedral blocks. Applying Proposition 3.12 inside $S_4$ also gives a purely analytic, weaker constant in the same corollary.

6.5. **Small primes and the remaining intersection orders.** The values $c_q$ can be computed exactly for small $q$. Every eigenspace of the graph $X_q$ in the proof of Theorem 6.6 is invariant under left translation by $G_q$, and each irreducible constituent of $\ell^2(G_q/A)$ has a non-zero $A$-fixed vector; hence every eigenvalue of $X_q$ already occurs on the functions on $A\backslash G_q/A$. Factoring the integer characteristic polynomial of this quotient gives

$$c_{11} = \frac{1+\sqrt{3}}{4}, \qquad c_{13} = \frac{3}{4}, \qquad c_{19} = \frac{1+\sqrt{5}}{4}.$$

| $q$ | 11 | 13 | 19 | 29 | 37 |
|---|---|---|---|---|---|
| All-pairs maximum | 0.683013 | 0.750000 | 0.809017 | 0.854361 | 0.845651 |

TABLE 2. Numerical all-pairs maxima, to be compared with $\sqrt{3}/2 \approx 0.866025$.

For pairs with $|A \cap B| < 6$ no identification with an arithmetic operator is available at present. Table 2 records, for five primes, the maximum of $\|P^{\rho}_{A} P^{\rho}_{gAg^{-1}}\|$ over all $g \notin A$ and all non-trivial irreducible $\rho$, computed in floating point from the operators induced on $A\backslash G_q/A$ by the double cosets $AgA$. In each case the maximum is attained by the pairs with $|A \cap gAg^{-1}| = 6$ and agrees with $c_q$ to the precision shown. Since an exterior octahedral subgroup is maximal, any two distinct ones generate $G_q$, so these are the quantities in Conjecture 1.2.

*Remark* 6.9 (Questions). By Proposition 6.3, Conjecture 1.2 concerns the pairs with intersection of order 1, 2, 3 or 4. A weaker question is whether the qualitative gap of Corollary 7.4, which holds for all generating exterior pairs outside a set of primes of relative density zero, holds for every prime $q \equiv \pm 3 \pmod 8$.

## 7. Uniform rates and the order of the mixing depth

The dichotomy supplies generating sets at bounded depth. Diameter estimates then control their contraction uniformly over all valid configurations. In particular, the unobstructed exterior classes of $\mathrm{PGL}_2(q)$ have base depth 2 or 3, uniformly in $q$; the issue here is the strength of contraction as the field grows. All logarithms in this section are natural unless their base is displayed. For a tree $\tau$, the law $\mu_\tau$ is unconditional.

**Theorem 7.1** (Diameter-to-mixing transfer)**.** *Consider a family of pairs $(G, C)$ satisfying case* (2) *of Theorem 3.7, with exact levels $D \le D_0$. Suppose there are constants $K, a > 0$, independent of the pair, such that every symmetric generating set $S$ of $G$ satisfies* $\operatorname{diam}(G, S) \le K(\log|G|)^a$. *Put $c = (8 \cdot 16^{D_0} K^2)^{-1}$. Every valid tree $\tau$ of level $m \ge D$ satisfies*

$$d_{\mathrm{TV}}(\mu_\tau, U_G) \le \frac{\sqrt{|G| - 1}}{2} \exp\left(-\frac{c 4^m}{(\log|G|)^{2a}}\right). \tag{7.1}$$

*For $D_T$, add the conditioning probability $p$ from Lemma 2.5. The leaves need not be involutions.*

*Proof.* Every sub-block of a valid tree is valid. Its level-$D$ sub-blocks therefore generate $G$ by Theorem 3.7. Each has $N = 4^D$ expanded occurrences, including inverse occurrences. Fix a non-trivial irreducible $\rho$. For each level-$D$ sub-block $\tau_0$, Proposition 3.12 gives

$$\|M_{\tau_0}(\rho)\| \le 1 - \frac{1}{8 \cdot 4^D K^2 (\log|G|)^{2a}}.$$

Expanding the recursion down to this level produces $4^{m-D}$ factors, counting adjoints, all with that norm bound. Submultiplicativity and $1 - x \le e^{-x}$ yield

$$\|M_\tau(\rho)\| \le \exp\left(-\frac{4^m}{8 \cdot 16^D K^2 (\log|G|)^{2a}}\right).$$

Apply Diaconis–Shahshahani as in Proposition 3.8, and use $D \le D_0$. □

For $0 < \eta < 1/2$, a sufficient condition for error at most $\eta$ is $m \geq D$ and

$$4^m \geq c^{-1}(\log|G|)^{2a}\left(\tfrac{1}{2}\log(|G|-1) + \log\frac{1}{2\eta}\right). \tag{7.2}$$

Thus fixed error, or error $|G|^{-b}$ for fixed $b > 0$, requires only $O(\log\log|G|)$ depth and $O((\log|G|)^{2a+1})$ expanded length.

**Proposition 7.2** (Support lower bound and optimal order)**.** *For any level-$m$ independent-read tree $\tau$,*

$$d_{\mathrm{TV}}(\mu_\tau, U_G) \geq \max\{0, 1 - 2^{4^m}/|G|\}.$$

*Consequently, for any fixed $0 < \eta < 1/2$, in an unbounded family satisfying Theorem 7.1, the least depth at which every valid tree has error at most $\eta$ is $\Theta(\log\log|G|)$, uniformly over the family.*

*Proof.* There are $4^m$ independent input bits, hence at most $2^{4^m}$ possible outputs. The complement of their support has uniform mass at least $1 - 2^{4^m}/|G|$ and output mass zero. Error at most $\eta$ therefore requires $4^m \geq \log_2((1-\eta)|G|)$. Valid trees exist at every level by Lemma 3.5, since $G$ is non-solvable. This rules out vacuous small levels. The upper bound is (7.2). □

The upper and lower bounds match in their order of depth, up to constant factors. On the obstructed side, the subgroup $H < G$ in Theorem 3.7(1) supplies at every level a valid tree $\tau$ supported in $H$, and hence $d_{\mathrm{TV}}(\mu_\tau, U_G) \geq 1 - |H|/|G| \geq 1/2$.

**Corollary 7.3** (Rank one and minimal simple groups)**.** *The uniform rate (7.1) and the optimal order of depth in Proposition 7.2 apply to:*

*(1) the unobstructed involution classes of $\mathrm{PSL}_2(q)$ and exterior classes of $\mathrm{PGL}_2(q)$ in Theorem 5.2;*

*(2) all finite minimal simple groups with any non-trivial conjugacy class, uniformly over both the group and the class.*

*Proof.* Pyber–Szabó's bounded-rank diameter theorem [53, Theorem 2 of the preprint] supplies constants $K, a$ for every symmetric generating set of a simple group of Lie type of bounded rank. For $G = \mathrm{PGL}_2(q)$, take a symmetric generating set $S$ and $t \in S$ outside $L = \mathrm{PSL}_2(q)$. With transversal $\{1, t\}$, Schreier generators of $L$ have length at most 3 in $S$; their inverses satisfy the same bound. Writing this symmetric set as $S_L$ gives

$$\mathrm{diam}(G, S) \leq 3\,\mathrm{diam}(L, S_L) + 1.$$

Thus the same type of diameter bound holds for $G$, with adjusted absolute constants. The rank-one levels are at most 3. For (2), Thompson's list in Corollary 4.3 has bounded Lie rank (at most 2, including the underlying type of the Suzuki groups), and Corollary 4.3 gives $D \leq 5$. Apply Theorem 7.1 with $D_0 = 3$ or 5, respectively, and then Proposition 7.2. □

7.1. **Uniform expansion outside exceptional primes.** Over prime fields, strong uniform expansion gives constant gaps for every generating block of bounded depth. The prime-field input goes back to Breuillard–Gamburd [14, Corollary 1.1]. Becker–Breuillard [8, Theorem 1.1 and Corollary 3.2] extend uniform expansion to bounded-rank families with bounded field degree, outside exceptional sets of characteristics.

**Corollary 7.4** (Uniform gaps for almost all primes)**.** *There is a set $\mathcal{E}$ of relative density zero among the primes such that the following hold for every prime $p \geq 5$ outside $\mathcal{E}$.*

*(1) For each fixed $K \geq 1$ there is $a_K > 0$, independent of $p$, such that every involutive block $\tau$ of level $k \leq K$ whose leaves generate $G = \mathrm{PSL}_2(p)$ or $\mathrm{PGL}_2(p)$ satisfies $\|M_\tau(\rho)\| \leq 1 - a_K$ for every non-trivial irreducible $\rho$.*

*(2) Let $C$ be an unobstructed involution class from Theorem 5.2: the exterior class in $\mathrm{PGL}_2(p)$, or the involution class in an unobstructed $\mathrm{PSL}_2(p)$. There is an absolute $c > 0$ such that every valid tree $\tau$ of level $m \geq D$ over $C$ satisfies*

$$\|M_\tau(\rho)\| \leq e^{-c4^m}, \qquad d_{\mathrm{TV}}(\mu_\tau, U_G) \leq \frac{\sqrt{|G|-1}}{2} e^{-c4^m}. \tag{7.3}$$

*The operator bound holds for every non-trivial irreducible $\rho$.*

*(3) If $p \equiv \pm 3 \pmod 8$, every pair of exterior subgroups $A, B \cong S_4$ generating $\mathrm{PGL}_2(p)$ satisfies $\|P_A^\rho P_B^\rho\| \leq 1 - b$ for every non-trivial irreducible $\rho$, with an absolute $b > 0$ independent of the pair and of $p$.*

*Proof.* Strong uniform expansion supplies $\varepsilon \in (0, 1]$ and $\mathcal{E}$ such that $|SX| \geq (1+\varepsilon)|X|$ whenever $|X| \leq |G|/2$ and $S$ is a symmetric generating set containing $e$, for both groups $G$. Starting with [14, Corollary 1.1], pass from $\mathrm{SL}_2(p)$ to its quotient $\mathrm{PSL}_2(p)$ and then to its index-two overgroup $\mathrm{PGL}_2(p)$; uniform expansion is preserved under these operations [8, Lemma 3.9]. Both sources use vertex expansion: the proof of [14, Corollary 1.1] produces a single $s \in S$ with $|sX \setminus X| \geq c|X|$, and [8, (1.1)] states the displayed form directly.

For (1), fix such a block $\tau$ and list its $r = 2^k$ leaf positions with their multiplicities. Their lazy walk assigns each distinct non-identity leaf probability at least $1/(2r)$. The expansion inequality gives at least $\varepsilon|X|$ outgoing edges from any $X$ with $|X| \leq |G|/2$, so its conductance is at least $\varepsilon/(2r)$. Cheeger's inequality gives spectral gap $\delta \geq \varepsilon^2/(8r^2)$, including these multiplicities. Each leaf occurs $2^k$ times in the expanded involutive block, so this is also its expanded-leaf walk. Put $N = 4^k = r^2$. By Proposition 3.14,

$$1 - \|M_\tau(\rho)\|^2 \geq \frac{\varepsilon^2}{4N(N+1)}.$$

Thus one may take $a_K = \varepsilon^2/(8 \cdot 4^K(4^K+1))$.

For (2), $D \leq 3$, and every level-$D$ sub-block of a valid tree $\tau$ generates $G$ by Theorem 3.7. Applying (1) at this fixed base level and expanding the recursion gives

$$\|M_\tau(\rho)\| \leq (1-a_3)^{4^{m-D}} \leq e^{-(a_3/64)4^m}.$$

Take $c = a_3/64$ and apply the Diaconis–Shahshahani bound.

For (3), fix a non-trivial irreducible $\rho$ on $V$, and write $P_A = P_A^\rho$, $P_B = P_B^\rho$. The walk $\nu = (U_A + U_B)/2$ has generating support $A \cup B$ and assigns each support element probability at least $1/48$. The same conductance argument gives spectral gap at least $\eta = \varepsilon^2/(2 \cdot 48^2)$. Its Fourier operator is $(P_A + P_B)/2$, which is positive semidefinite. For a unit vector $v \in V^A$, $1 + \|P_B v\|^2 \le 2(1-\eta)$. Hence $\|P_A P_B\| \le \sqrt{1-2\eta} \le 1-\eta$; if $V^A = 0$ the norm is zero. Take $b = \eta$. □

Part (3) is a qualitative form of Conjecture 1.2 outside $\mathcal{E}$: a gap for all generating exterior pairs, without the sharp constant of Theorem 6.6.

### 7.2. **Few exceptional configurations over arbitrary fields.**

**Corollary 7.5** (Exceptional generating blocks at fixed depth)**.** *Fix integers $R, k \ge 1$ and $\delta > 0$. There is $a = a(R,k,\delta) > 0$ such that, for every finite simple group $G(q)$ of Lie type of rank at most $R$, all but at most $O_{R,k,\delta}(q^\delta)$ simultaneous $G(q)$-conjugacy classes of generating involutive level-$k$ configurations $\tau$ satisfy*

$$\|M_\tau(\rho)\| \le 1 - a \qquad \text{for every non-trivial irreducible } \rho.$$

*The same conclusion holds for $\mathrm{PGL}_2(q)$, $q \ge 4$. Restricting to valid configurations, or to specified conjugacy classes of involution leaves, preserves this absolute bound on exceptions.*

*Proof.* The dimension-zero theorem of Becker–Breuillard [8, Theorems 1.2 and 4.1], with $2^k$ entries, gives an expansion constant $\varepsilon = \varepsilon(R,k,\delta) > 0$ outside the stated number of simultaneous conjugacy classes of generating tuples. The finite simple groups have the required bounded-dimensional algebraic embeddings; $\mathrm{PGL}_2(q)$ is covered by Theorem 4.1 in the adjoint group of type $A_1$, since it contains $\mathrm{PSL}_2(q)$ and therefore satisfies the density hypothesis of that theorem for $q$ large. Finitely many small fields are absorbed into the implied constant. For an involutive tuple put $N = 4^k$. Its lazy leaf walk, with multiplicities, has gap at least $\varepsilon^2/(8N)$ by the conductance argument in Corollary 7.4. By Proposition 3.14 one may take $a = \varepsilon^2/(8N(N+1))$, after reducing $\varepsilon$ to at most 1. Conjugating every leaf conjugates the block operator, so the transfer respects exactly the equivalence relation counted by the source theorem. □

Here the depth is fixed and the field degree is unrestricted. The exception count is absolute; a relative-density statement in a restricted family additionally requires a lower bound on its size.

*Remark* 7.6 (A question). Do the involution classes of the minimal simple families $\mathrm{PSL}_2(2^\ell)$, $\mathrm{PSL}_2(3^\ell)$ and $\mathrm{Sz}(2^\ell)$ admit a constant gap for every valid configuration of level $D$, uniformly as the prime exponent $\ell$ grows? Corollary 7.5 allows exceptional configurations; removing them in these families would follow from the bounded-rank uniform expansion conjecture discussed in [8, Introduction].

## Declarations

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**Declaration of competing interest.** The author declares no competing interests.

**Data availability.** The verification artefact, its outputs and the extended version of this paper are archived at Zenodo [61]. The repository

https://github.com/ernest-zermeno/commutator-tree-mixing-finite-groups

contains the same material. The README of the artefact identifies exact certificates, numerical checks, sampled runs and the certificate taken from [63].

**Mathematical software.** Python and NumPy were used for the computations described in Section 2.2; SymPy was used to factor integer characteristic polynomials and isolate roots in the exact octahedral certificates.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used OpenAI GPT Astra, Anthropic Claude Opus 5.5 and Google Gemini 3.1 Pro to search for counterexamples and errors in proofs, assist with developing and verifying code, survey related literature, edit the language and translate the author's arguments into English. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article, including the mathematical statements, proofs, references and code.

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